Succinct Structure Representations for Efficient Query Optimization

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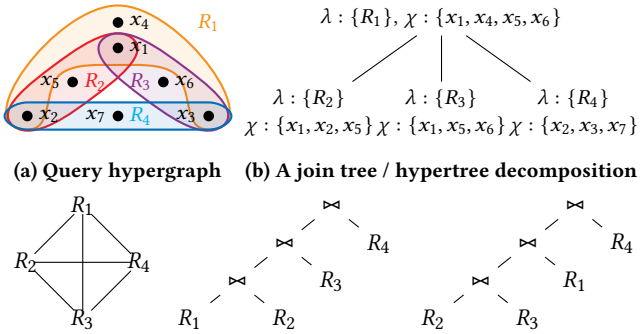
Abstract

Structural decomposition methods for conjunctive queries offer powerful theoretical guarantees for join evaluation, yet they are rarely used in real-world query optimizers. A major reason is the difficulty of combining cost-based plan search and structure-based evaluation. In this work, we bridge this gap by integrating structural properties of queries into cost-based optimization. To achieve this, we introduce meta-decompositions for acyclic queries, a novel representation that succinctly represents all possible join trees and enables their efficient enumeration. Meta-decompositions can be constructed in polynomial time and have sizes linear in the query size. Building on this, we design an efficient polynomial-time cost-based optimizer based directly on the meta-decomposition, without the need to explicitly enumerate all possible join trees. We characterize plans found by this approach using a novel notion of width, which effectively connects the join tree structure and thus the theoretical guarantees of running time to any query plan. Experimental results demonstrate that the plans in our class are consistently comparable to those found by the state-of-the-art dynamic programming approach, while our planning process runs orders of magnitude faster, especially on large and complex queries.

1 Introduction

Joins are the cornerstone of relational database queries. However, query optimizers face extreme difficulty when planning large multi-way join queries. The search space of join orders grows exponentially as the number of relations increases. In fact, finding the optimal join order of a query is well-known to be NP-hard in general [11, 34]. For this reason, beyond a handful of joins (e.g., more than 12), current query optimizers must abandon exhaustive dynamic programming and resort to heuristic or greedy algorithms. As a result, real-world optimizers often miss the best plans and produce suboptimal ones that can be orders of magnitude slower.

Nevertheless, the database theory community has long been providing tools that exploit query structures to tackle this complexity. The *Yannakakis algorithm* [44] is one classic example that utilizes join trees of acyclic queries to obtain an instance-optimal guarantee on the asymptotic complexity for evaluating these queries. Hypertree decompositions [18] and related width measures [3, 4] generalize this approach to provide similar strong worst-case optimal guarantees of evaluation time. Over the years, these structure-based methods have been further generalized to evaluate joins with projections [5, 13, 24], aggregations [2, 26], differences [25, 47], top-k [41], comparisons [42], etc., all with strong theoretical performance guarantees. These results suggest that leveraging a query's structural properties could be of great help for the search for query plans and could dramatically improve evaluation performance with robust worst-case bounds. Yet, in practice, despite limited attempts by, e.g., Scarcello et al. [33] and RelationalAI [3], query optimizers in real-world systems rarely utilize such structure-based approaches; therefore, they continue to struggle with large, complex queries.



(a) Query hypergraph **(b) A join tree / hypertree decomposition**

(c) Query graph **(d) A width-1 query plan** **(e) A width-2 query plan**

Figure 1: The hypergraph, a join tree, the query graph, and query plans of the query in Example 1.1

Why are structural methods hardly adopted in mainstream systems? A core obstacle is the dichotomy between cost-based plan search and structure-based evaluation. Traditional cost-based optimizers usually find query plans with the optimal or near-optimal costs under some cost model using (1) **exact algorithms**, mostly based on dynamic programming, such as [30, 35, 36, 38, 39], which have to run in exponential time, thus not scalable to very large numbers of relations; (2) **greedy strategies** based on heuristics such as greedy operator ordering (GOO) [9, 15, 31], which have guarantees of neither optimality nor a reasonable approximation factor unless under some assumption, such as independence of predicate selectivities, of the cost function; and/or (3) **transformation-based techniques**, such as the Volcano/Cascades framework [20, 21] and genetic algorithms [7]. Although certain approaches leverage the structures of, e.g., *query graphs* [9, 30, 31], they do not necessarily correspond to any structural decompositions. Therefore, join orders generated by these methods may not necessarily correspond to any optimal structural decomposition. This makes it impossible to directly apply structure-based methods to them.

Example 1.1. Consider the query

$$Q = \pi_\emptyset(R_1[x_1, x_2, x_3, x_4] \bowtie R_2[x_1, x_2, x_5] \bowtie R_3[x_1, x_3, x_6] \bowtie R_4[x_2, x_3, x_7])$$

This is a Boolean acyclic query. Its hypergraph is shown in Fig. 1a and a join tree, which is also a hypertree decomposition, in Fig. 1b. Its query graph, as shown in Fig. 1c, is a clique with the four relations as vertices, as each pair shares some join attribute. Fig. 1d is a query plan that follows the structure of the join tree (formally defined later). Fig. 1e does not, but it is still within the search space of traditional algorithms according to the query graph.

On the other hand, theoretical research typically focuses on the asymptotic data complexity, assuming a given decomposition is available. But in practice, there can be an exponentially large number of possible decompositions for a given query, and explicitly

scanning through all of them would be computationally prohibitive. Choosing the right decomposition is therefore difficult without a cost-based exploration of alternatives. Therefore, despite the appealing asymptotic running-time guarantee, Yannakakis-style algorithms have very limited adoption in practice due to their considerable hidden overhead.

1.1 Our Contributions

In this work, we take a major step towards bridging structural decomposition techniques with cost-based query optimization. We **introduce a novel *width* measurement for query plans** and show that, for query plans of width 1, we can effectively embed the query's tree structure within a traditional query plan, providing guidance for query planning even in the absence of cost information. In particular, we give an example that the evaluation of acyclic Boolean conjunctive queries is in *instance-optimal time* using width-1 plans. We also present a method which, for any given join tree of a query, constructs the optimal width-1 query plan that adheres to the join tree in $O(|Q|)$, *linear in the query size* $|Q|$.

By searching for plans of minimal width, we restrict the search space to only those that follow the query's optimal tree decompositions. To make this practical, we introduce *meta-decompositions* for acyclic conjunctive queries that **universally encode, in one structure, the entire family of join trees for the query**. Despite the possibly exponential number of join trees of a given query, this representation **can be constructed in polynomial time and has size linear in the query size**. We give an efficient procedure that enumerates all join trees without repetition from a meta-decomposition, as a potential interface for other structure-based techniques that require explicit enumeration of the join tree.

Nevertheless, explicitly enumerating all possible join trees still incurs an exponential overhead. Therefore, we develop an efficient ***structure-guided cost-based optimization* framework to find the optimal width-1 query plan among all join trees** without explicitly enumerating all candidates. We prove that if the query's meta-decomposition has bounded fan-out, then *the global optimal width-1 query plan can be found in linear time* $O(|Q|)$. For general queries, we provide efficient heuristics that leverage meta-decomposition to guide the optimizer towards good query plans without requiring exhaustive search.

Although we currently restrict our focus to acyclic queries only, such queries in fact account for the overwhelming majority of real workloads [10, 28]: among 8,125 queries across popular benchmarks, 7,929 (97.59 %) are acyclic (See Appendix A.1 for detailed breakdowns). Optimizing acyclic queries has therefore been of sustained interest by the community [6, 8, 25, 40–43, 48]. However, current approaches typically run outside the cost-based optimization framework of real systems. Our approach provides a tool to close this gap by integrating cost-based reasoning into structure-based approaches.

We then empirically demonstrate that our approach can indeed be game-changing for query optimization for large joins. We **implement an optimizer metaDecomp** and evaluate it on queries from 4 benchmarks, which include many large, complex queries. We compare the results against the native optimizer of DuckDB and a state-of-the-art dynamic programming algorithm, DPconv [36]. The results show that width-1 plans consistently match or even outperform the optimal plans found by these baselines. Moreover, by exploiting query structures, our optimization process is orders of magnitude faster for large queries. We also demonstrate that join tree enumeration can be orders-of-magnitude faster than existing approaches based on naïve GYO reductions, and that selecting join trees using our cost-based optimization framework significantly improves the effectiveness of existing structure-based methods.

Paper organization. The rest of the paper is organized as follows: After laying out the necessary background in Section 2, we first define the *width* notion of query plans and discuss the connection between query structures and the widths of query plans in Section 3. In Section 4, we define *meta-decompositions*, the algorithm to construct them, as well as the way to enumerate all join trees based on them. In Section 5, we demonstrate how to perform cost-based optimization using meta-decompositions. In Section 6, we present experimental results to show the efficiency of query plans generated by this approach. We conclude and discuss possible future work in Section 7. Due to the space constraints, all missing proofs can be found in the appendix.

2 Preliminaries

We write 2^S for the set of all subsets of a set S. If $f : A \to B$ and $S \subseteq A$, we let $f(S) = \{f(a) \mid a \in S\}$. For a set of sets S, we write $\cup S = \bigcup_{x \in S} x$. If $f : A \to T$, where T is a set, and $S \subseteq A$, we let $f(S) = \bigcup_{x \in S} f(x)$.

2.1 Conjunctive Queries and Hypergraphs

Conjunctive queries can be represented by hypergraphs, and our terminology follows the seminal references on this topic [16–19, 40]. Let $\mathcal{D} = \{R_1[\bar{x}_1], \cdots, R_n[\bar{x}_n]\}$ be a database consisting of n relations, where $\bar{x}_i$ is the set of attributes of relation R_i. A conjunctive query (with self-predicates) is defined as

$$Q \leftarrow \pi_{\bar{y}} \left(\boldsymbol{\sigma_1}(R_1[\bar{x}_1]) \bowtie \cdots \bowtie \boldsymbol{\sigma_n}(R_n[\bar{x}_n])\right),$$

where $\boldsymbol{\sigma_i}$ is the set of selection predicates involving only attributes in $\bar{x}_i$, $\bar{y}$ is the set of output attributes, and $\bowtie$ denotes natural join (with attribute renaming to avoid ambiguity). If $\bar{y} = \cup_{i \in [n]} \bar{x}_i$, the query is a full join query and we omit $\bar{y}$ for simplicity; if $\bar{y} = \emptyset$, the query is a boolean query that returns true if there exists a tuple satisfying all specified joins and predicates, or false otherwise. We also require the query to be *self-join-free* at the logical level: If multiple relations correspond to the same physical relation, they are treated as distinct logical relations with unique renamings.

We define the associated *hypergraph* for the conjunctive query Q as $H(Q) = (V(H), E(H))$, where the vertices set $V(H)$ consists of all attributes appearing in the query, i.e., $V(H) = \cup_{i \in [n]} \bar{x}_i$, and the *hyperedges* set $E(H) \subseteq 2^{V(H)} \setminus \{\emptyset\}$ contains one hyperedge for each relation, i.e., $E(H) = \{e_1, \cdots, e_n\}$, where $e_i = \bar{x}_i$.

The *primal graph* of a hypergraph H is the *graph* on $V(H)$ which is obtained from H by turning each hyperedge of H into a clique. A hypergraph is connected if its primal graph is connected. *All hypergraphs in this paper are assumed to be finite, connected, and to have a non-empty set of hyperedges.*

2.2 Acyclicity and Join Trees

A conjunctive query is acyclic if its associated hypergraph is acyclic, and we define the notion of acyclicity of a hypergraph in the standard way in database theory, e.g., as in [1, Section 6.4]. A hypergraph H is acyclic if and only if the output of the GYO algorithm [22, 45] on H is empty [1, Theorem 6.4.5]. Additionally, a conjunctive query Q is acyclic if and only if there exists a valid width-1 hypertree decomposition, e.g., a valid *join tree* for the hypergraph $H(Q)$ [1, Theorem 6.4.5].

Slightly different from the usual definition of join trees [1], where each vertex is simply a relation, in this paper, we define them using notations for hypertree decompositions [18], to make it easier to generalize to the meta-decompositions that we will define later.

Let Q be an acyclic conjunctive query with associated hypergraph H. A *join tree* for Q is a triple (T, χ, λ) where T is a tree, $\chi : V(T) \to 2^{V(H)}$ is a function s.t. $\chi(V(T)) = \bigcup_{p \in V(T)} \chi(p) = V(H)$, and $\lambda : V(T) \to E(H)$[1], and it satisfies the following conditions:

- (H_1) **Coverage condition.** For each $e \in E(H)$, there exists exactly one $p \in V(T)$ with $\lambda(p) = \{e\}$.
- (H_2) **Connectedness condition.** For each $v \in V(H)$, the set of vertices $\{p \in V(T) \mid v \in \chi(p)\}$ induces a connected subtree of T. Or, equivalently, for all pairs of vertices $p, q \in V(T)$ and all s on the (unique) path between p and q on T, we have $\chi(p) \cap \chi(q) \subseteq \chi(s)$.
- (H_3) **Width-1.** For all $p \in V(T)$, $|\lambda(p)| = 1$ and $\chi(p) = V(\lambda(p))$.

Example 2.1. Let's revisit the query discussed in Example 1.1. Its hypergraph is illustrated in Fig. 1a. It is easy to verify that the join tree in Fig. 1b satisfies conditions (H_1) and (H_3). Regarding condition (H_2), consider, e.g., attribute x_1. The vertices in the join tree that include x_1 are the ones with R_1, R_2, and R_3, which induces a connected subtree. Similar arguments apply for all attributes.

We assume that all trees are rooted. For two trees T_1, T_2 of $H(Q)$, rooted at r_1 and r_2 respectively, with $E(T_1) = E(T_2)$, we still consider them to be different trees if $r_1 \neq r_2$. When writing "$(p, q) \in E(T)$" in a tree T, we refer to the edge $\{p, q\} \in E(T)$ with p being closer to the root than q. Moreover, we write T_p as the subtree rooted at p, i.e., T_p contains all vertices q such that p is on the unique path from the root of T to q. For every neighbor q of vertex p in T, we write $T_{p \to q}$ for the subtree with vertices

$$V(T_{p \to q}) = \{s \in V(T) \mid \text{the path from } p \text{ to } s \neq p \text{ in } T \text{ contains } q\}.$$

We also define the *fan-out* of any $p \in V(T)$ as

$$f(p) = \begin{cases} \deg(p) - 1, & \text{if } p \text{ is not the root} \\ \deg(p), & \text{if } p \text{ is the root} \end{cases}$$

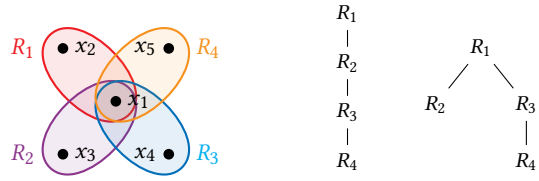
where $\deg(p)$ is the degree (number of neighbours) of node p, and $f(p)$ corresponds to the number of children of p on T. The *fan-out* of the tree T is the maximum fan-out among all nodes

$$f(T) = \max_{p \in V(T)} f(p).$$

Given a join tree T, we recursively define the *induced query* for every $p \in V(T)$ as

$$Q(p) = \begin{cases} p & \text{if } p \text{ is a leaf node} \\ p \bowtie \left(\bowtie_{c \text{ is a child node of } p} Q(c) \right) & \text{Otherwise.} \end{cases}$$

[1]Due to space limitations, we may simply use hyperedges (e.g., R_1) to label the vertices on join trees instead of writing out the full λ and χ labels when the context is clear.



(a) Hypergraph of the query **(b) Two valid join trees**

Figure 2: Hypergraph and two valid join trees of the star query Q in Example 2.3

One important observation is that *an acyclic conjunctive query may have an exponential number of distinct join trees.*

Theorem 2.2. *Consider* star queries[2] *of the form*

$$Q_{\text{star}} \leftarrow R_1[\bar{x}_1] \bowtie \cdots \bowtie R_n[\bar{x}_n],$$

where for any $i, j \in [n]$, $\bar{x}_i \cap \bar{x}_j = \{x\}$, for some attribute x. There are n^{n-1} possible join trees for such a star query with n relations.

Example 2.3. Consider the query $Q = R_1[x_1, x_2] \bowtie R_2[x_1, x_3] \bowtie R_3[x_1, x_4] \bowtie R_4[x_1, x_5]$. Q is a star query of the form we consider. Its associated hypergraph is as shown in Fig. 2a, and Fig. 2b shows two of its valid join trees.

In fact, the possible join trees of such queries in Theorem 2.2 and Example 2.3 are simply *all possible trees with n vertices* (or, equivalently, all spanning trees of a complete graph K_n), each labeled by one relation. Any such tree satisfies the connectedness condition (H_2). There are a total of n^{n-2} distinct non-rooted trees, and each tree has n possible root nodes. Each such tree can be encoded by a *Prüfer sequence* [32] of length $n - 2$ (where each element in the sequence can take values between 1 and n for a tree with n vertices), plus one extra indicator for the root node. This gives a way to enumerate such trees without repetition.

2.3 Query Plans and Join Ordering

A *query plan* $\mathcal{P} = (V(\mathcal{P}), E(\mathcal{P}))$ for a query Q is a rooted tree, where each non-leaf node is a relational operator (e.g, join, projection, selection), and each leaf node corresponds to a base relation. A plan specifies an order of operations to evaluate the query, where the outputs of child nodes serves as inputs to the parent. For the purpose of join ordering, *we assume selections and projections are inherently integrated in the join operation at the lowest possible level, and therefore we use only join operators in internal nodes.*

Given a query plan $\mathcal{P}$, we recursively define the *induced query* for every $p \in V(\mathcal{P})$ as

$$Q(p) = \begin{cases} p & \text{if } p \text{ is a leaf node,} \\ Q(c_1) \bowtie Q(c_2) & \text{if } p \text{ is join operation.} \end{cases}$$

In addition, we exclude Cartesian products. For every join node with two child nodes c_1 and c_2, inducing subqueries $Q(c_1)$ and $Q(c_2)$, with output attributes $\bar{y}_1$ and $\bar{y}_2$, respectively, we require $\bar{y}_1 \cap \bar{y}_2 \neq \emptyset$.

In this work, we focus on query optimization of binary joins, where, for every join operation on the query plan, there are exactly two child nodes. Given a conjunctive query with more than 2 rela-

[2]We note that, in some literature, e.g., [9], such queries are called "*clique*" queries, because their *query graphs* are cliques. Regardless, it has been shown that optimizing such queries exactly is NP-hard and takes exponential time [9, 34] (unless P = NP).

tions, the query plan also specifies the sequence and structure in which joins are to be evaluated. A query plan is called *left-deep* (or *right-deep*) if all join operations have their right (left) child node corresponding to a base relation; otherwise, the query plan is referred to as *bushy*.

We apply a cost function C to select the best query plan, where C can be represented as a recursive function along the query plan $\mathcal{P}$ and root node r, as follows:

$$C(\mathcal{P}) = \begin{cases} c(r), & \text{if } r \text{ is single relation} \\ c(r) \oplus C(\mathcal{P}_1) \oplus C(\mathcal{P}_2), & \text{if } r \text{ is join operation} \end{cases}$$

where $c(r)$ is the cardinality of the output of r, and $\mathcal{P}_1$ and $\mathcal{P}_2$ (if applicable) are the left and right subtrees of r, respectively. In this work, we adopt the cost function C_{out} to minimize the intermediate result size by substituting the $\oplus$ operator simply with arithmetic +, as is standard in, e.g., [36].

The standard approach to find an optimal plan is to recursively find the optimal split of a set of relations S into two disjoint subsets S_1 and S_2 with $S = S_1 \cup S_2$ [30, 35, 36, 38, 39]. With a given cost function C, the optimization problem can be solved by dynamic programming (DP) with a worst-case time complexity $O(3^n)$ given by the algorithm DPccp [30]. The recently proposed algorithm DPconv [36] improves this computation by utilizing fast subset convolution, reducing the complexity for this cost model C_{out} to $O(2^n n^3 W \log(Wn))$, where W is the largest join cardinality.

3 Widths of Query Plans & Width-1 Plans

We start by introducing the width notion for a given query plan P, which connects the plan's structure to the potential size of its intermediate results. This measure helps quantify the complexity of a query plan before looking at the specific database instance. Given a query plan $\mathcal{P}(V(\mathcal{P}), E(\mathcal{P}))$, for any node $p \in V(\mathcal{P})$, we consider the subquery evaluated at that node. Let $H_p = (V(H_p), E(H_p))$ be the hypergraph induced by the leaf nodes in the subtree rooted at p with the induced query $Q(p) = \bowtie_{e \in E(H_p)} e$. Let $\overline{H_p} = (V(\overline{H_p}), E(\overline{H_p}))$ be the residual hypergraph corresponding to the rest of the query, where $E(\overline{H_p}) = E(H) \setminus E(H_p)$, and $V(\overline{H_p}) = \bigcup_{e \in E(\overline{H_p})} e$.

The set of attributes that separates H_p and $\overline{H_p}$ is the interface $I(p) = V(H_p) \cap V(\overline{H_p})$. We then define

$$Q'(p) \leftarrow \pi_{I(p)} Q(p),$$

which captures the minimal set of attributes that need to be kept in the intermediate result at node p, because this intermediate result must then be joined with the results from the rest of the query. The set of join attributes is precisely the interface $I(p)$. The width of node p is the size of the smallest subset of its relations needed to cover this interface, i.e.,

$$w(p) = \min \left\{ |S| : S \subseteq E(H_p) \text{ and } I(p) \subseteq \cup S \right\}.$$

and the width of P is the maximum width over all nodes, i.e.,

$$w(P) = \max_{p \in V(\mathcal{P})} w(p).$$

The width of a query plan indicates an upper bound on the minimal intermediate results over all nodes p during evaluation.

Theorem 3.1. *Given a Boolean conjunctive query Q and a query plan $\mathcal{P}$ for the query,*

$$\max_{p \in V(H)} |Q'(p)|$$

is upper bounded by $O(N^{w(\mathcal{P})})$ in the worst case, where N is the maximum cardinality of any base relation.

Such a width notation can draw a direct connection with the join tree, as follows.

Definition 3.2. We say a query plan $\mathcal{P} = (V(\mathcal{P}), E(\mathcal{P}))$ is *induced by* a join tree $T = (V(T), E(T))$ if

(1) for each node $p \in V(T)$, there exists $q \in V(\mathcal{P})$ with $p = Q(q)$, or $Q(p) = Q(q)$, i.e., the induced query of node p in the join tree is exactly the induced query of node q in the query plan, unless $Q(q)$ involves only a single relation; and
(2) for each non-leaf node $q \in V(\mathcal{P})$ with two children u and v, there exists some $s \in E(H_u)$ and $t \in E(H_v)$ such that $I(u) \subseteq s$, $I(v) \subseteq t$, and there exists an edge between s and t in $E(T)$.

Theorem 3.3. *Given a query plan $\mathcal{P}(V(\mathcal{P}), E(\mathcal{P}))$ for query Q, the width $w(\mathcal{P}) = 1$ if and only if there exists a join tree $T = (V(T), E(T))$ for Q that induces the query plan $\mathcal{P}$.*

Theorem 3.3 directly lead to the relationship between width-1 query plans and acyclic queries:

Theorem 3.4. *A conjunctive query Q has width-1 query plans if and only if Q is acyclic.*

Example 3.5. Consider the query

$$Q = R_1[x_1, x_2, x_3] \bowtie R_2[x_1, x_4, x_5] \bowtie R_3[x_5, x_6] \bowtie R_4[x_3, x_7].$$

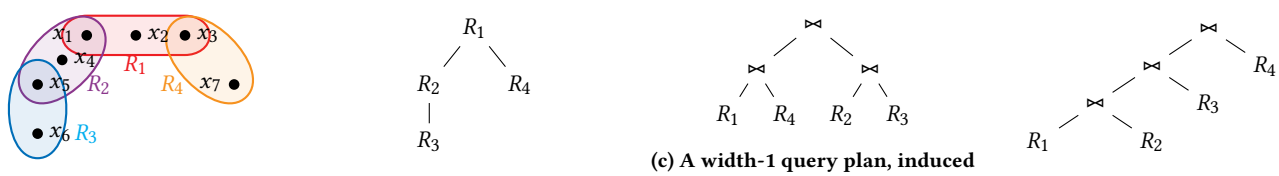
Its hypergraph is shown in Fig. 3a and a join tree in Fig. 3b. Given this join tree, Fig. 3c is an example of a valid width-1 query plan, whereas Fig. 3d shows a query plan that is width-2: taking p as the node on the join tree with R_2, there is no q on the query plan such that $Q(p) = Q(q) = R_2 \bowtie R_3$.

The width-1 query plan plays an important role in tree-based query evaluation methods. Tree-based query evaluation methods follow a common pattern: given a join tree, they apply a series of reductions starting from the leaves and working upwards until only a single node remains. The selected join tree heavily influences the efficiency of the execution plan produced by these methods. In essence, width-1 query plans exploit the join tree structure, which evaluates the query in a bottom-up fashion, where each node $p \in V(T)$ completes all its internal joins with its child subtrees before joining with its parent. Such a strategy provides several benefits.

Early elimination of irrelevant attributes. By the connectedness condition of join trees, attributes that are not part of the output but only participate in the subtree can be projected out in an early stage, as long as the parent no longer contains the attribute.

Example 3.6. With the query in Example 3.5, assuming that x_3 is not an output attribute, if we choose the width-2 query plan in Fig. 3d, x_3 has to be kept in all intermediate results before R_4 is joined at the very end, whereas, in the width-1 query plan in Fig. 3c, it can already be projected out after completing $R_1 \bowtie R_4$.

Local optimization. Instead of searching in a vast space of arbitrary join orders for all relations at once, we can locally optimize the join order at each node of the tree. For each node $p \in V(T)$



(a) Hypergraph of the query **(b) A valid join tree** **(c) A width-1 query plan, induced by the join tree shown in Fig. 3b** **(d) A width-2 query plan**

Figure 3: The hypergraph, a join tree, a width-1 query plan, and a width-2 query plan of the query in Example 3.5

with children $c_1, \cdots, c_{f(p)}$ in the join tree, we only need to determine the optimal join order between p and $Q(c_1), \ldots, Q(c_{f(p)})$. The order of doing these joins can be optimized independently for each node, without considering other parts of the query. Such an approach, therefore, significantly prunes the search space for optimization, making plan selection much more tractable, and can yield near-optimal performance for practical queries. Our experimental results show that the maximum fan-out $f(T)$ tends to be very low. In the queries we experimented with in the join ordering benchmark (JOB), the maximum is 5, and the average is 3.

Example 3.7. Consider again the join tree in Fig. 3b for the query in Example 3.5. Given this join tree, a width-1 query plan requires that a subtree with $R_2 \bowtie R_3$ exist in the query plan. It therefore remains only to optimize the order to join R_1, $(R_2 \bowtie R_3)$, and R_4, which is the local optimization problem at the root node R_1 of the join tree. Suppose that the cardinalities $|R_1 \bowtie R_4| < |(R_2 \bowtie R_3) \bowtie R_1|$. Then, the optimal plan in our cost model is the one in Fig. 3c.

Theorem 3.8. *Given a join tree T, the optimal width-1 query plan induced by T can be found in time $O\left(f(T)2^{f(T)}|Q|\right)$.*

Admittedly, this approach comes at the cost of excluding some plan alternatives with higher widths. However, those excluded plans are not compatible with structure-guided algorithms and will not preserve the theoretical guarantees by the query structure.

Structure-guided query evaluation. An additional advantage of width-1 query plans is that they naturally accommodate those newly developed structure-guided query evaluation methods. For example, we can simulate the Yannakakis algorithm for evaluating *relation-dominated* conjunctive queries [40] by following a width-1 query plan. A conjunctive query is relation-dominated if it is acyclic and there exists a relation $R_i(\bar{x}_i)$ that contains all the output attributes, i.e., $\bar{y} \subseteq \bar{x}_i$. It shall be noted that an acyclic Boolean query is also a relation-dominated query, since $\bar{y} = \emptyset$.

Theorem 3.9. *A relation-dominated conjunctive query can be evaluated in $O(|db|)$ time using a width-1 query plan, where $|db|$ represents the size of the database.*

To create a linear-time width-1 query plan for evaluating a relation-dominated query, let R_i denote the relation containing all output attributes. One can construct the query plan using any join tree with R_i as the root. The width-1 query plan will then evaluate the query in a bottom-up fashion. Because all output attributes are located at the root of the join tree and at the end of the query plan, when the query plan is evaluating node p, all attributes in T_p but not in p are removed from the query, guaranteeing a linear running time.

Example 3.10. Consider again the query in Example 1.1, along with the hypergraph, a join tree, and two query plans, as shown in Fig. 1. Assume that it is evaluated over a database instance with $R_1 = \{(i, i, i, i) \mid 1 \leq i \leq n\}$, $R_2 = R_3 = R_4 = \{(i, j, j) \mid 1 \leq i, j \leq n\}$ for some $n > 1$. The query plan shown in Fig. 1d has width 1, as the interface of the node p that induces the query $Q(p) = R_1 \bowtie R_2$ is $I(p) = \{x_1, x_2, x_3\}$, which can be covered by only one hyperedge, R_1. The case is similar for the subquery $(R_1 \bowtie R_2) \bowtie R_3$, whose interface $\{x_2, x_3\}$ can be covered by R_1. In this case, *any width-1 query plan can give a linear runtime*, as we can start from R_1, apply a series of semi-joins, and the intermediate sizes will always be at most the size of $|R_1|$. On the contrary, the query plan in Fig. 1e has width 2 because the interface of q that induces the query $Q(q) = R_2 \bowtie R_3$ is $I(q) = \{x_1, x_2, x_3\}$, which cannot be covered by any single hyperedge $e \in \{R_2, R_3\}$. *This would cause a quadratic blowup of the intermediate result size*, as each tuple $(i, j, j) \in R_2$ can be joined with n tuples (i, j', j') with $1 \leq j' \leq n$ from R_3.

To conclude this section, we highlight some important distinctions between our definition of width-1 query plans and Yannakakis-style query plans [40, 44]. Although they are both guided by join trees, width-1 query plans do not incorporate full reductions using semi-joins as is done in the Yannakakis algorithm. We have shown, nevertheless, that these plans help in early elimination of irrelevant attributes and still preserve favourable theoretical guarantees of the asymptotic complexity for Boolean conjunctive queries (Theorem 3.1) and relation-dominated queries (Theorem 3.9). In Section 6, we will experimentally show that these plans, in practice, are indeed highly efficient for most queries. Furthermore, without the need for full reductions, width-1 query plans are much easier to integrate into the query execution frameworks of current database systems, allowing for more efficient and easier cost-based optimization, as we will discuss in the remainder of the paper.

4 Meta-Decompositions

Although we have shown a connection between join trees and width-1 query plans in Theorem 3.1 and Theorem 3.3, it is impractical to find the optimal width-1 query plan by enumerating every possible join tree—As noted earlier, a query can have an exponential number of possible join trees. Therefore, in this section, we introduce and explore a generalized representation that allows us to efficiently represent all join trees of an acyclic query. This enables us to perform query optimization directly on this representation, eliminating the need to enumerate all possible join trees.

We highlight an observation that *the exponential number of possible join trees often arises when multiple relations share common join attributes*, as is the case for star queries discussed in Theorem 2.2 and illustrated by Example 2.3. In such queries, any pair of relations

in Q_{star} has valid join predicates, meaning there are $\binom{n}{2} = \frac{n(n-1)}{2}$ pairwise predicates. Any specific join tree selects only $n-1$ of those predicates and infers the remaining predicates through the tree structure, where $\frac{(n-1)(n-2)}{2}$ predicates are not selected for that particular tree. On the contrary, in the case of a query where, for any attribute x, there are at most two relations $R_i(\bar{x}_i)$ and $R_j(\bar{x}_j)$ such that $x \in \bar{x}_i$ and $x \in \bar{x}_j$, as in, e.g., Example 3.5, the query demonstrates a more "tree-like" structure compared to Q_{star}. There is a unique tree structure for such a query, with n distinct, structurally isomorphic join trees, differing only by the choice of the root relation.

In light of this, we introduce *meta-decompositions*. In the case of acyclic conjunctive queries, it can capture all possible join trees but requires only linear space.

Definition 4.1. Given a conjunctive query Q and the associated hyergraph $H(Q)$, its *meta-decomposition* $M = (V(M), r(M), E(M), \lambda, \chi, \kappa)$ is a labeled tree, where

- $(V(M), r(M), E(M))$ is a tree with vertices $V(M)$, a unique root $r(M) \in V(M)$, and edges $E(M)$,
- $\lambda : V(M) \to 2^{E(H)}$ maps each tree vertex to either a set of hyperedges in H or to $\emptyset$, and
- $\chi : V(M) \to 2^{V(H)}$ and $\kappa : V(M) \to 2^{V(H)}$ each maps each tree vertex in M to a set of vertices in H,

such that it satisfies (H_1), (H_2), and

(H_3') For all $p \in V(M)$, $\chi(p) \subseteq V(\lambda(p))$ if $\lambda(p) \neq \emptyset$.

(H_4) **Interface condition.** For every non-root node $p \in V(M)$, let q be p's parent node, $V(M)$ be the set of vertices in M, $\kappa(p)$ satisfies
  (a) $\kappa(p) = \chi(p) \cap \chi(V(M) \setminus V(M_p)) \subseteq \chi(q)$, where M_p is the subtree of M rooted at p; and
  (b) $\kappa(p) \nsubseteq \chi(s)$ for all $s \in V(M) \setminus V(M_q)$ if $\kappa(p) \neq \chi(q)$.
  For root node $p = r(M)$, we set $\kappa(p) = \emptyset$.

(H_5) **Uniqueness condition.** For every pair of nodes $p, q \in V(M)$ such that $\kappa(p) = \kappa(q)$, there exists one unique node $o \in V(M)$ with $\lambda(o) = \emptyset$ and $\chi(o) = \kappa(p) = \kappa(q)$.

For acyclic queries, the meta-decompositions additionally satisfy

(H_3'') For all $p \in V(M)$ such that $\lambda(p) \neq \emptyset$, we have $|\lambda(p)| = 1$ and $\chi(p) = V(\lambda(p))$.

The meta-decomposition differs from the standard join tree in the following ways. First, we introduce a κ-label using the interface condition (H_4): Given a meta-decomposition M, for any node $p \in V(M)$, $\kappa(p)$ captures the *interface* between the induced hypergraph $H(V(T_p))$ and the complementary hypergraph $H \setminus H(V(T_p))$. Removing hyperedge $\lambda(p)$ from the query hypergraph H partitions the hypergraph into multiple subhypergraphs, each intersecting with $\lambda(p)$ precisely at these interfaces, where for p, the interfaces include $\kappa(p)$ and $\kappa(c)$ for every child node c of p in M. Condition (H_4)(b) limits the shape of the meta-decomposition M: there can be multiple nodes that can satisfies $\kappa(p) \subseteq \chi(q)$, while Condition ($H_4$)(b) limits q to be the highest node with $\kappa(p) \subseteq \chi(q)$.

Furthermore, we introduce nodes with $\lambda = \emptyset$. We refer to such nodes as *minor nodes* and to other nodes as *physical nodes* in the meta-decomposition. Minor nodes serve exactly to represent scenarios where multiple relations share a common interface, as captured by condition (H_5). When several relations have a common interface

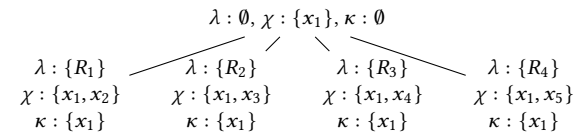


Figure 4: Meta-decomposition of the query in Example 2.3

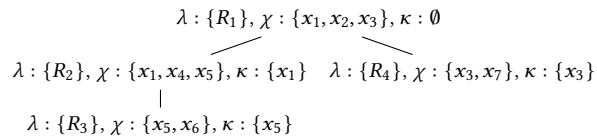


Figure 5: Meta-decomposition of the query in Example 3.5

k, as for the example of a star query, let p be a relation that contains the interface with $k \subseteq \chi(p)$, we can either place them in the same subtree of p on the join tree or separate them into different subtrees. The use of minor nodes facilitates this flexibility without the need to introduce an exponential number of edges in the representation.

Example 4.2. The meta-decomposition of the query in Example 2.3 is shown in Fig. 4. The root of this tree is a minor node, showing that the four children share precisely the attribute x_1. Note also that the κ-label of all 4 non-root nodes is exactly $\{x_1\}$.

Example 4.3. The meta-decomposition of the query in Example 3.5 is shown in Fig. 5. It has the exact same structure as the join tree shown in Fig. 3b and has no minor node.

Example 4.4. Consider the query

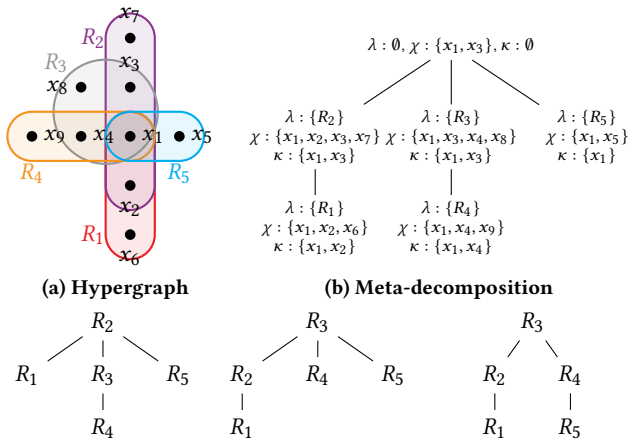
$$Q = R_1[x_1, x_2, x_6] \bowtie R_2[x_1, x_2, x_3, x_7] \bowtie R_3[x_1, x_3, x_4, x_8] \bowtie R_4[x_1, x_4, x_9] \bowtie R_5[x_1, x_5]).$$

Its hypergraph and meta-decomposition are shown in Fig. 6a and Fig. 6b, respectively. Fig. 6c and Fig. 6d show two valid join trees of this hypergraph, which should be intuitive based on the meta-decomposition. We also highlight the fact that R_5 could in fact be the child of any other node on the join tree, because its only interface with the rest of the hypergraph is $\{x_1\}$. For example, Fig. 6e is also a valid join tree.

4.1 Constructing Meta-Decompositions

A meta-decomposition can be constructed by Algorithm 1. In the algorithm, $o(e, H) = e \cap \left(\cup_{e' \in E(H) \setminus \{e\}} e'\right)$ denotes the interface, as a set of vertices, of hyperedge e with the rest of the hyperedges $E(H) \setminus \{e\}$ in the hypergraph H. A hyperedge e is *reducible* if there exists another hyperedge e', such that $o(e, H) \subseteq e'$. We refer to such hyperedges as *ears*. Similar to the GYO algorithm, the algorithm is also based on reduction. The GYO algorithm removes one ear for each round, thereby establishing a total order for all hyperedges in order to identify a specific join tree. In contrast, Algorithm 1 *removes all ears in every round* until only a single hyperedge remains in the hypergraph. This process introduces only a partial order for all hyperedges, which *maintains the flexibility to represent all possible ordering of ear removal, and hence all possible join trees*, through the resulting meta-decomposition.

For every round, the algorithm first handles the case that some edges e_1, e_2 are *mutually reducible*, i.e., $o(e_1, H) = o(e_2, H)$ (Lines 4-12). The algorithm will create a minor node p with $\chi(p) = o(e_1, H)$



(a) Hypergraph **(b) Meta-decomposition**

(c) A join tree **(d) A second join tree** **(e) A third join tree**

Figure 6: Hypergraph, meta-decomposition, and three join trees of the query in Example 4.4

if such a node does not exist, and then remove e_1 and e_2 from the hypergraph. We then add a hyperedge $e_m = o(e_1, H)$, which conceptually corresponds to the minor node, back to the hypergraph, so that the acyclicity of H is not broken. The rationale for adding e_v can be illustrated by the following example:

Example 4.5. Consider the query

$$Q = \pi_\emptyset(R_1[x_1, x_2, x_3, x_4] \bowtie R_2[x_1, x_2, x_5] \bowtie R_3[x_1, x_3, x_6] \bowtie R_4[x_2, x_3, x_6] \bowtie R_5[x_1, x_2, x_3, x_7]),$$

which is similar to Example 3.10 but contains an additional relation R_5. In this case, at the beginning, R_1 and R_5 are both ears and mutually reducible, as $o(R_1, H) = o(R_5, H) = \{x_1, x_2, x_3\}$. However, if we simply remove both R_1 and R_5 at the same time, *the remaining hyperedges, R_2, R_3, and R_4, form a cyclic hypergraph.* But, if we add $e_m = o(R_1, H) = o(R_5, H) = \{x_1, x_2, x_3\}$, then e_v, R_2, R_3, and R_4 still form an acyclic query, and the algorithm can proceed to pick R_2, R_3, and R_4 as ears, before e_m is picked and removed at the very end.

We formalize this idea as follows:

Lemma 4.6. *Given an acyclic hypergraph H, if there exist two distinct ears $e_1, e_2 \in E(H)$ such that $o(e_1, H) = o(e_2, H)$, then, the hypergraph H', with $E(H') = E(H) \setminus \{e_1, e_2\} \cup \{o(e_1, H)\}$ and $V(H') = \cup E(H')$, is acyclic.*

Proof sketch. In the GYO algorithm for H, since $o(e_1, H) = o(e_2, H) \subseteq e_2$, e_1 is an ear and can already be removed, with e_2 being its witness. So, there exists a join tree T of H where there is a leaf node p_1 with $\lambda(p_1) = \{e_1\}$, and its parent p_2 has $\lambda(p_2) = \{e_2\}$. If we merge p_1 and p_2 into a single node p_m with $\lambda(p_m) = \{o(e_1, H)\}$ to obtain a new tree T', T' is a join tree of H'. □

In each round, the algorithm then finds and removes the set $\mathcal{E}$ of all ears of H'. For each $e \in \mathcal{E}$, we create a physical node p and set $\kappa(p) = o(e, H')$. During the procedure, the algorithm creates minor nodes if necessary (Lines 9, 14, 17–18, and 20).

Theorem 4.7. *Algorithm 1 terminates in time $O(|E(H)|^3)$.*

Algorithm 1: Constructing the meta-decomposition of an acyclic hypergraph

input : An acyclic hypergraph H
output : The meta-decomposition M of H

1 $H' \leftarrow$ a hypergraph with $V(H') = V(H)$ and $E(H') = E(H)$
2 $V(M) \leftarrow \emptyset$
3 **while** $|E(H')| > 1$ **do**
4   **foreach** *maximal $S \subseteq E(H')$ such that $|S| > 1$, and, for any $e_1, e_2 \in S$, $o(e_1, H') = o(e_2, H')$* **do**
5     $o \leftarrow$ the value of $o(e_i, H')$ that all $e_i \in S$ share
6   **if** *there exists $v \in V(M)$ such that $\lambda(v) = \emptyset$ and $\chi(v) = o$* **then** // There is already a minor node. We remove it and create a new one later
7   Remove v from $V(M)$
8   **foreach** $e \in S$ **do**
9   Add a node p to $V(M)$ with $\lambda(p) = \{e\}$ if $e \in E(H)$, or $\lambda(p) = \emptyset$ otherwise, $\chi(p) = e$, and $\kappa(p) = o(e, H')$
10   Remove e from $E(H')$
11   Add a special hyperedge $e_m = o$ to $E(H')$
12   $\mathcal{E} \leftarrow$ the set of all ears $e \in E(H')$, i.e., for every $e \in \mathcal{E}$, there exists a distinct $e' \in E(H')$ with $o(e, H') \subseteq e'$
13   **foreach** $e \in \mathcal{E}$ **do**
14   Add a new node p to $V(M)$ with $\lambda(p) = \{e\}$ if $e \in E(H)$, or $\lambda(p) = \emptyset$ otherwise, $\chi(p) = e$, and $\kappa(p) = o(e, H')$
15   Remove e from $E(H')$
16     $S \leftarrow \{p' \in V(M) \cup \mathcal{E} \mid \kappa(p') = \kappa(p)\}$
17   **if** $|S| > 1$ *and, for all $p' \in S$, $\lambda(p') \neq \emptyset$* **then**
18   Add a new node v to $V(M)$ with $\lambda(v) = \emptyset$, $\chi(v) = \kappa(p)$, and $\kappa(v) = \kappa(p)$
19 $e \leftarrow$ the remaining edge in $E(H')$
20 Add a new node r to $V(M)$ with with $\lambda(r) = \{e\}$ if $e \in E(H)$, or $\lambda(r) = \emptyset$ otherwise, $\chi(r) = e$, and $\kappa(r) = \emptyset$
21 Remove e from $E(H')$
22 $E(M) \leftarrow \emptyset$
23 **foreach** $p \in V(M)$ *such that $\kappa(p) \neq \emptyset$* **do**
24   **if** *there exists $q \in V(M)$ such that $\lambda(q) = \emptyset$ and $\chi(q) = \kappa(p)$* **then**
25   Add an edge (q, p) in $E(M)$
26   **else**
27   Find $q \in V(M)$ such that $\kappa(p) \subseteq \chi(q)$ and $\kappa(p) \not\subseteq \kappa(q)$
28   Add an edge (q, p) in $E(M)$
29 **return** $M = (V(M), r, E(M), \lambda, \chi, \kappa)$

Proof sketch. The key observations are (1) that the size $|E(H')|$ decreases by at least one in each iteration of the while-loop (Line 3), and (2) that, in each iteration of the while loop, all sets S (Line 4) can be enumerated in $O(|E(H')|) = O(|E(H)|)$ time, if we maintain a hash map that maps each $o \subseteq V(H')$ to all hyperedges $e \in E(H')$ such that $o(e, H') = o$, which is updated whenever a hyperedge e is added or removed from $E(H')$. □

Theorem 4.8. *Given an acyclic query with an acyclic hypergraph H, Algorithm 1 returns a valid meta-decomposition of H.*

Theorem 4.9. *In the meta-decomposition M of an acyclic hypergraph H as constructed by Algorithm 1, the number of vertices $|V(M)|$ and the number of edges $|E(M)|$ are $O(|E(H)|)$.*

Proof sketch. Vertices $V(M)$ in M are either physical nodes or minor nodes. We create one physical node per hyperedge in $E(H)$. And for minor nodes, we use the fact that each minor node m has fan-out $f(m) \geq 2$. □

4.2 Join Tree Enumeration

The meta-decomposition allows one to efficiently enumerate all possible join trees of a query, as we will show by an algorithm in this section. This not only serves as proof of the universality of the meta-decomposition but also provides a powerful approach to enumerating join trees, which could be of great help for other structure-based methods of query optimization.

As an intuition, we can observe that the distinct join trees of the same hypergraph can be different in the following ways:

(1) **"Re-branching".** If a vertex p in a join tree has children c_1 and c_2 such that $\kappa(c_1) = \chi(c_1) \cap \chi(p) \subseteq \chi(c_2)$, then c_1 could have been a child of c_2 without violating any condition. In the meta-decomposition, condition (H_4) ensures that c_1 is positioned at the highest possible node. Therefore, when enumerating a join tree, we can make c_1 a child of c_2 if $\kappa(c_1) \subseteq \chi(c_2)$. The join tree Fig. 6e based on meta-decomposition Fig. 6b from Example 4.4 demonstrates such a case.
(2) **"Rerooting".** A rerooting of a join tree is still a valid join tree.
(3) **Minor nodes.** If multiple relations share the same set of attributes pairwise, they can be joined in arbitrary order. This situation is represented by the minor nodes in the meta-decomposition. An example is the root node with χ-label $\{x_1\}$ in Fig. 4 for Example 2.3, based upon which we should be able to enumerate, among others, the join trees in Fig. 2b.

These alternatives should be considered to enumerate all join trees, and the meta-decomposition retains all the necessary information to facilitate this. Algorithm 2 is an algorithm to enumerate all join trees. For each node in the meta-decomposition starting from the root, it recursively enumerates all possible join trees given by the subtree rooted at each of its children, and then enumerates all possible ways to combine these join trees into one. For minor nodes r, we find its neighbours that share the common set of attributes given by $\chi(r)$ and generate all possible distinct structures of join trees without repetition (using the Prüfer sequence), where each vertex will be replaced by the join tree for a subtree given by a neighbour, and at this point we consider all rerootings of some tree as identical. At the very end of this algorithm, we return all the combined join trees we obtain, and it remains only to collect all the rerootings. Due to the space constraints, the algorithms for enumerating all trees and rerootings are given in Appendix C.1.

Example 4.10. Take the query in Example 4.4 as an example. Its meta-decomposition is shown in Fig. 6b. Let the root node be r, and we use $v_1, \ldots v_5$ to denote the nodes such that $\lambda(v_i) = \{R_i\}$. r is a minor node, with a $\chi(r) = \{x_1, x_3\}$. We start by enumerating all possible join trees induced by the subtrees of the meta-decomposition,

Algorithm 2: Enumerating all join trees based on a meta-decomposition

input : A meta-decomposition $M = (V(M), r, E(M), \lambda, \chi, \kappa)$
output : All possible non-homomorphic join trees up to rerooting

1 **Function** enumRec(v: *a node on* M):
2 $C \leftarrow$ set of children c of v on M, sorted by partial order $\supseteq$ on $\kappa(c)$
3 **if** $\lambda(v) = \emptyset$ **then** // minor node
4 $S \leftarrow \{c \in C \mid \kappa(c) = \chi(v)\}$ // they can be found at the head of the sorted C
5 **foreach** $c \in S$ **do**
6 $\mathcal{T}_c \leftarrow$ enumerate(c) // All non-homomorphic join trees for subtree M_c rooted at c, up to rerooting
7 **if** $\kappa(v) = \chi(v)$ **then**
8 $\mathcal{T}_P \leftarrow$ enumerateTrees($S \cup \{v\}$), all rerooted to v // All possible trees with vertices $S \cup \{v\}$, by Prüfer sequences // The minor node v will eventually be replaced on Line 33
9 **else** // $\kappa(v) \neq \chi(v)$, the parent should not participate
10 $\mathcal{T}_P \leftarrow$ enumerateTrees(S)
11 **foreach** $T_P \in \mathcal{T}_P$ **do** // A skeleton of a way to connect subtrees, given by a Prüfer sequence
12 **foreach** $c \in V(T_P)$ **do**
13 **if** *c is a leaf node* **then**
14 $\mathcal{T}'_c \leftarrow \mathcal{T}_c$
15 **else**
16 $\mathcal{T}'_c \leftarrow \emptyset$
17 **foreach** *non-leaf $c \in V(T_P)$, in bottom-up order, $T_c \in \mathcal{T}_c$, child d of c on T_P, $T_d \in \bigcup_{T_d \in \mathcal{T}_d}$* rerootings$(T_d, \emptyset, \kappa(d))$, *$u \in V(T_c)$ such that $\kappa(d) \subseteq \chi(u)$* **do**
18 $T'_c \leftarrow T_c$, with T_c added as a child of u
19 Add T'_c to $\mathcal{T}'_c$
20 $\mathcal{T} \leftarrow \mathcal{T} \cup \mathcal{T}'_{\text{root}(T_P)}$
21 $X \leftarrow C \setminus S$, preserving the order
22 **else**
23 $\mathcal{T} \leftarrow$ a set of one tree with one vertex v only
24 $X \leftarrow C$, preserving the order
25 **foreach** *child $c \in X$ in order* **do**
26 $\mathcal{T}_c \leftarrow \bigcup_{T \in \text{enumerate}(c)}$ rerootings$(T, \emptyset, \kappa(c))$
27 $\mathcal{T}' \leftarrow \emptyset$
28 **foreach** $T_c \in \mathcal{T}_c, T \in \mathcal{T}$ **do** // Enumerate all possible v in some T which T_c can be attached to
29 **if** $\lambda(c) = \emptyset$ *and* $\kappa(c) = \chi(c)$ **then** // The root of T_c would be a minor node from Line 8. We find a parent of each of its children.
30 **foreach** *child d of the root of T_c* **do**
31 $\mathcal{T}'' \leftarrow \emptyset$
32 **foreach** $T' \in \mathcal{T}', u \in T'$ *such that* $\kappa(d) \subseteq \chi(u)$ **do**
33 $T'' \leftarrow T'$ with T_d attached as a child of u
34 Add T'' to $\mathcal{T}''$
35 $\mathcal{T}' \leftarrow \mathcal{T}''$
36 **else**
37 **foreach** $u \in T$ *such that* $\kappa(c) \subseteq \chi(u)$ **do**
38 $T' \leftarrow T$ with T_c added as a child of u
39 Add T' to $\mathcal{T}'$
40 $\mathcal{T} \leftarrow \mathcal{T}'$
41 **return** $\mathcal{T}$
42 **return** $\bigcup_{T \in \text{enumRec}(r)}$ rerootings$(T, \emptyset, \emptyset)$

rooted at each child of r. For the subtree with v_2 and v_1, there is only one valid join tree, rooted at v_2, which has one child v_1. It is not possible to reroot to v_1 because $\kappa(v_2) = \{x_1, x_3\} \not\subseteq \chi(v_1)$. The case is similar for the subtree with v_3 and v_4. And the subtree with v_5 induces only one join tree with one node v_5. We then enumerate all possible trees with vertices v_2 and v_3, having $\kappa(v_2) = \kappa(v_3) = \chi(r)$, using Prüfer sequences. There are only two such trees, each of which contains one edge connecting v_2 and v_3, and they are re-rootings of each other. For each tree, we replace v_2 and v_3 by their induced join tree. In the end, we have one child v_5 with $\kappa(v_5) \neq \chi(r)$. It can be attached as a child of any node v such that $\kappa(v_5) \subseteq \chi(v)$. In this example, it can, in fact, be a child of any other node. Fig. 6c is a join tree where the spanning tree generated by the Prüfer sequence is rooted at R_2, and R_5 is attached as a child of R_2. Fig. 6d is a similar example where the spanning tree is rooted at R_3, and R_5 is attached as a child of R_3. And Fig. 6e is based on the same spanning tree as for Fig. 6d, but R_5 is attached as a child of R_4 instead of R_3.

We now show that this algorithm is both correct (producing valid join trees) and complete (enumerating all possible join trees).

Theorem 4.11 (Correctness). *Given a meta-decomposition M of an acyclic hypergraph H, all trees T output by Algorithm 2 are valid join trees of H.*

Theorem 4.12 (Completeness). *Given a meta-decomposition M of an acyclic hypergraph H, Algorithm 2 enumerates all possible join trees of H.*

As is detailed in the proof of Theorem 4.12, for each join tree T, we can pinpoint exactly one way in which T can be enumerated by the algorithm based on the meta-decomposition M. Noting also that the number of children of each node on the meta-decomposition M is bounded by the fan-out $f(M)$, we get the following running time of Algorithm 2, following the control flow.

Theorem 4.13 (Running time). *Given a meta-decomposition M of an acyclic hypergraph H, Algorithm 2 enumerates all possible join trees of H in $O(f(M) \cdot |V(H)| \cdot |\mathcal{T}|)$ time, where $|\mathcal{T}|$ is the number of all possible join trees.*

In fact, this algorithm can be written in a way that enumerates join trees in polynomial delay, taking polynomial space. The details are given in Appendix C.8.

5 Cost-Based Optimization from Meta-Decompositions

Although we have presented an algorithm, enumerating all possible join trees is infeasible, as their number grows super-exponentially with the number of relations, as shown by Theorem 2.2. Instead, we propose a cost-based optimization algorithm that *operates directly on the meta-decomposition* to find an optimal width-1 query plan. Similar to the case for join tree enumeration discussed in Section 4.2, the search space for such plans is defined by four key factors: (1) selecting a root of the join tree to induce the width-1 query plan; (2) unnesting minor nodes; (3) handling re-branching possibilities, and (4) determining the local join order for each node and its neighbors. In this section, we detail the cost-based optimization procedure, which systematically addresses all four factors.

Algorithm 3: Find the optimal width-1 query plan to evaluate the query induced by a meta-decomposition

input : A meta-decomposition $M = (V, r, E)$
output : The optimal query plan

```
// Bottom-up traversal
foreach non-root node q ∈ V(M), in bottom-up order do
    p ← q's parent node on M
    plan(T_{p→q}) ← optimizeLocal(q, {plan(T_{q→c}) | c ∈ C(q)})   // C(q) is the set of all children of node q
plan(M) ← optimizeLocal(r, {plan(T_{r→c}) | c ∈ C(r)})
// Top-down traversal
foreach c ∈ C(r) do
    plan(T_{c→r}) ← optimizeLocal(r, {plan(T_{r→c'}) | c' ∈ C(r) \ {c}})
foreach non-root node q ∈ V(M), in top-down order do
    p ← q's parent node on M
    foreach child c ∈ C(q) do
        plan(T_{c→q}) ← optimizeLocal(q, {plan(T_{q→p})} ∪ {plan(T_{q→c'}) | c' ∈ C(r) \ {c}})
return the plan plan(T_{p→q}) ⋈ plan(T_{q→p}) with the minimum cost among all (p, q) ∈ E(M)
```

5.1 Re-Branching

As previously discussed, re-branching can happen when a child node c of a node p in the meta-decomposition could also be a child of another descendant node d of p in a join tree, if $\kappa(c) \subsetneq \kappa(d)$. This scenario, however, is in fact very rare in practice: we found no queries in the JOB benchmark that exhibited this scenario, and only one query in the TPC-H benchmark can involve re-branching. Therefore, for simplicity, we do not consider re-branching when describing the algorithm in this section. As we will show in the appendix, accommodating a possible re-branching adds only a constant factor to the overall complexity.

5.2 The Overall Algorithm

Given a meta-decomposition, our algorithm to produce a width-1 query plan is shown in Algorithm 3. This algorithm makes two traversals of the meta-decomposition M, one bottom-up and one top-down. In the bottom-up traversal, for each node q on the meta-decomposition, we find the optimal plan to join the relation in q and all the plans given by the children of q, which gives the optimal plan to evaluate $Q(T_q)$, the induced query on the subtree of M rooted at q. Then, the top-down traversal, for each node q and each child c of q on the meta-decomposition, we find the optimal plan to join q and all plans given by the neighbors (parent and children) of q except c, which gives the optimal plan to join all relations in $T_{c\to q}$. At the end, we choose the root of the query plan, which is the minimum-cost plan of the form $\mathsf{plan}(T_{p\to q}) \bowtie \mathsf{plan}(T_{q\to p})$ among all $(p, q) \in E(M)$. Alternatively, we can think of this algorithm as *finding the optimal rerooting* of M, so that the query is evaluated by (1) replacing each node with a join with its children and (2) removing its children, in a bottom-up order.

The function optimizeLocal called in the algorithm optimizes the local join for each node and its neighbors. We will elaborate on this in Section 5.3.

5.3 Optimizing the Local Join with neighbors

One essential component of this algorithm is to, for a vertex q in the meta-decomposition, find the optimal plan to join the relation in this vertex and the optimal query plans given by (a subset of) its neighbors (optimizeLocal). This can be viewed as a classic star join optimization problem, where R_q is the hub relation, and the result of the induced query $Q(T_{c_i})$ from each of its neighbors c_i is a satellite relation, denoted by R_i for $i = 1, \ldots, |C_q|$, where $\chi(c_i) \cap \chi(q) \neq \emptyset$ and $\chi(c_i) \cap \chi(c_j) \subseteq \chi(q)$ for all $i \neq j$. Unfortunately, even in this restricted setting, this problem of optimizing the join of R_q and all R_i's is still NP-hard, because this can be reduced from the problem of optimizing the ordering a Cartesian product $S_1 \times S_2 \times \cdots \times S_n$—which is already shown to be NP-hard [34]—by adding a shared attribute to each S_i such that all tuples have the same value for this attribute. By using dynamic programming algorithms [30, 36], the problem can be solved in $\widetilde{O}\left(2^{|C_q|}\right)$, where $\widetilde{O}$ suppresses polylogarithmic factors and $|C_q|$ equals the fan-out of q on the meta-decomposition. However, as mentioned in Section 3, *the fan-out of real-world queries is typically very small.* This makes the exponential-time exact algorithm practical for the vast majority of cases.

In the rare case that a node has a high fan-out, making the exact DP algorithm too slow, we can substitute it with a heuristic. An effective choice is Greedy Operator Ordering (GOO) [15], which iteratively joins the neighbor that results in the smallest intermediate result and results in a left-deep plan. This method is outlined in Algorithm 8 in Appendix D.2.

5.4 Case Study: Optimizing Relation-Dominated Queries with Meta-Decompositions

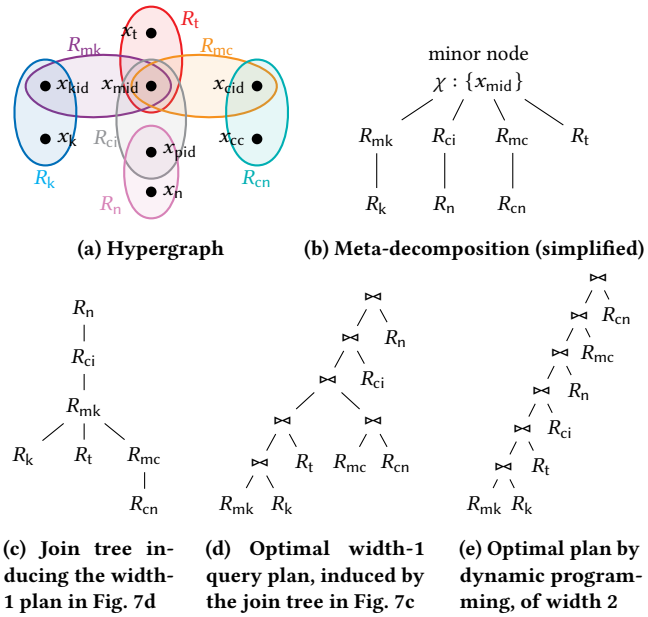
Our framework naturally incorporates projection pushdown for early cardinality reduction. When building a plan for a subquery, such as $\mathsf{plan}(T_{p\to q})$ in Algorithm 3, we can immediately project the result to only the attributes that are necessary for subsequent operations. Specifically, after computing the joins within the subtree T_q, we only need to retain (1) attributes required for the final output $(O \cup \chi(T_{p\to q})) \cup (\chi(p) \cap \chi(q))$, and (2) attributes required for joining with the remaining relations, which is $\kappa(q)$ by definition.

This strategy is especially effective for relation-dominated queries, where there exists one relation that contains all output attributes. If a subtree contains no output attributes, we can project its result to only $\kappa(q)$, i.e., the join keys with the remaining relations, which can significantly reduce the size of intermediate results. However, traditional dynamic-programming-based optimizers often struggle to identify these opportunities for projection pushdown, as they lack the comprehensive structural view provided by the meta-decompositions and the width-1 query plans.

Example 5.1. Take query 17f in the join order benchmark (JOB) as an example. Omitting irrelevant attributes in each relation, this query can be abstractly represented as

$$\begin{aligned} Q \leftarrow \pi_{x_\mathsf{n}}(&R_\mathsf{ci}[x_\mathsf{mid}, x_\mathsf{pid}] \bowtie R_\mathsf{cn}[x_\mathsf{cid}, x_\mathsf{cc}] \bowtie \sigma_{x_\mathsf{k}=\cdots}(R_\mathsf{k}[x_\mathsf{kid}, x_\mathsf{k}]) \\ &\bowtie R_\mathsf{mc}[x_\mathsf{mid}, x_\mathsf{cid}] \bowtie R_\mathsf{mk}[x_\mathsf{mid}, x_\mathsf{kid}] \\ &\bowtie \sigma_{x_\mathsf{n}=\cdots}(R_\mathsf{n}[x_\mathsf{pid}, x_\mathsf{n}]) \bowtie R_\mathsf{t}[x_\mathsf{mid}, x_\mathsf{t}]). \end{aligned}$$

This is a relation-dominated query, as it has only one output attribute x_n, which is an attribute of R_n. Fig. 7 shows the hypergraph,



(a) Hypergraph (b) Meta-decomposition (simplified) (c) Join tree inducing the width-1 plan in Fig. 7d (d) Optimal width-1 query plan, induced by the join tree in Fig. 7c (e) Optimal plan by dynamic programming, of width 2

Figure 7: The hypergraph, the meta-decomposition, the optimal width-1 query plan with the inducing join tree, and the optimal query plan (of width 2) found by dynamic programming, of the query JOB 17f in Example 5.1

the optimal width-1 query plan (found by metaDecomp given cardinalities of the real data set) along with the join tree inducing it, and a width-2 query plan (which is the plan given by dynamic programming). In the width-1 plan in Fig. 7d, *all intermediate join results can be projected to a single attribute*, since the interface is always only one attribute, and there is no output attribute before the final join with R_n. But, with the width-2 query plan in Fig. 7e, *the result of* $((R_\mathsf{mk} \bowtie R_\mathsf{k}) \bowtie R_\mathsf{t}) \bowtie R_\mathsf{ci}$ *has to be projected to three attributes,* x_mid, x_pid, *and* x_n—the former two are join keys, and x_n is the output attribute. Executing these plans, we indeed observe that *width-1 plan has a* 1.72x *speedup over the width-2 plan.*

Like evaluating join-projection queries, one can modify the cost model and utilize a join-tree-based query evaluation algorithm that incorporates cost-based optimization through meta-decomposition. We hope this work inspires further research in related areas and encourages the integration of these theoretically desirable query evaluation techniques in practical systems.

6 Experimental Evaluation & Analysis

In this section, we present experimental results to provide clear and insightful answers to the following research questions:

(RQ1) How fast is query optimization using meta-decompositions?
(RQ2) How good are width-1 plans, compared to the optimal?
(RQ3) To what extent can the proposed join tree enumeration techniques help structure-based query evaluation?

6.1 Experimental Setup

6.1.1 Experimental Environment. Experiments are conducted on an Apple M4 Pro machine with 14 cores and 48 GB RAM, running macOS Tahoe 16.0. We use DuckDB as our test database because of

its excellent performance and simple cost model. The experiments use DuckDB 1.2.2, Scala 3.6.3, GraalVM 17.0.12, and LLVM 20.1.3.

We implement metaDecomp in Scala. It first computes the meta-decomposition for a given query (Section 4) and then applies our cost-based optimization algorithm (Section 5) to select the best width-1 query plan. During the experiments, we compare the runtime of the query plans selected by (1) metaDecomp, (2) the state-of-the-art dynamic programming optimizer DPconv [36] , and (3) DuckDB's native optimizer, which uses DPccp [30] for queries with up to 12 relations and Greedy Operator Ordering (GOO) [15] otherwise. Query plans by metaDecomp and DPconv are enforced in DuckDB by rewriting queries into a sequence of subqueries that create temporary views.

Each query plan is executed 10 times, and we report the median optimization and execution time. To ensure accuracy, we exclude I/O time from runtime measurements. The full dataset is preloaded into the DuckDB database, and the estimated cardinalities are preloaded into memory, with all loading time excluded from optimization timing. We enforce a 5-minute timeout on execution time. Queries that none of the three optimizers can complete within this time limit are excluded from the results.

6.1.2 Queries, Benchmarks, and Cardinalities. We evaluate our approach on the following four benchmarks:

(1) The SPJ (select-project-join) queries in the **decision support benchmark (DSB)** [14]. We only consider acyclic queries (240 out of 300) and evaluate them on a 10 GB database instance.
(2) The **Join Order Benchmark (JOB)** [27] over the IMDB dataset. All queries in this benchmark are acyclic.
(3) **Musicbrainz**, a benchmark proposed by [29] with significantly larger queries that join up to 26 relations. We take only the acyclic queries from the benchmark (243 out of 375).
(4) **JOBLarge**, a new benchmark we introduce to extend JOB to simulate practical but larger analytical scenarios, where each query is generated by selecting two queries from the JOB that share at least one common output attribute and merging them into a single query, with a join on a shared attribute, such that the query remains acyclic. We only retain generated queries that take DuckDB at least 50 ms to evaluate. JOBLarge significantly increases query complexity, with queries with an average of 18, and maximum 34 relations, compared to the standard JOB (average of 8, maximum 17).

For the relatively small benchmarks DSB and JOB, we provide true cardinalities of possible intermediate join results to both metaDecomp and DPconv (precomputed by running appropriate `COUNT(*)` queries), in the approach and format consistent with the implementation of DPconv [36]. This provides DPconv with perfect information to identify the optimal plan, and enables metaDecomp to determine the optimal width-1 plan. For the larger benchmarks Musicbrainz and JOBLarge, computing exact cardinalities for all possible intermediate joins is infeasible due to the much larger query sizes. Therefore, we use estimated cardinalities instead, via DuckDB's `EXPLAIN` command. Note, however, that these estimates can sometimes be highly inaccurate and may not reflect the true relative sizes of intermediate results.

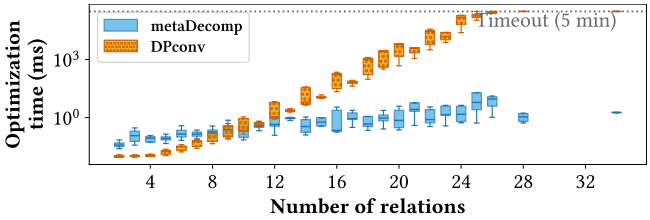


Figure 8: Box plot of optimization time for all 4 benchmarks

6.2 Results

Due to space constraints, we present summarized figures for all benchmarks. Detailed figures are included in Appendix E.

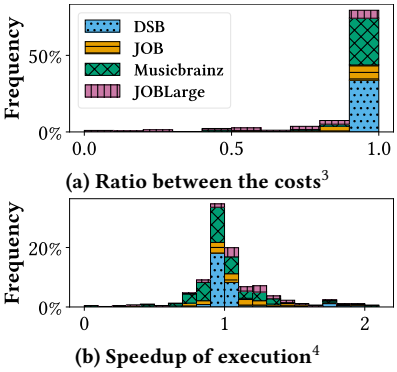
6.2.1 (RQ1) Optimization Speed Using metaDecomp. We first compare the time taken for optimization by metaDecomp and DPconv. The overall results are shown in Fig. 8. The figure clearly shows that the optimization time of metaDecomp remains almost constant as the number of relations increases, whereas DPconv exhibits an exponential growth trend, and its optimization time starts to surpass that of metaDecomp from approximately 9 relations. For many queries in Musicbrainz and JOBLarge (starting from approximately 25 relations), DPconv fails to find a valid plan within the 5-minute time limit, again highlighting that traditional exponential-time dynamic-programming-based optimizers do not scale to such large queries, whereas metaDecomp still takes merely milliseconds.

6.2.2 (RQ2) Quality of width-1 query plans. Even though metaDecomp restricts its search to width-1 plans, as shown in Fig. 9a, in most queries, it *finds query plans with cost values comparable to the optimal*. Furthermore, Fig. 9b shows the speedups achieved by executing the optimal width-1 plans over executing the globally optimal plan. Note that, although one would typically not expect speedups since the optimal plans have the minimum possible cost values, in many cases, the optimal width-1 query plans still exhibit speedups in the actual execution, thanks to the possibilities of early projections by exploiting the structural information of queries.

6.2.3 (RQ1) & (RQ2) Together: Overall Performance.

metaDecomp over DPconv. Taking into account both optimization and execution time, we can observe, in Table 1, Fig. 10a, and Fig. 11a, that metaDecomp shows a clear advantage overall. This is especially the case for queries that are more complex, for which DPconv takes more than 1 second (10^3 ms), as shown in the points towards the right in Fig. 11a, the major cause of which is the exponential optimization time. This demonstrates that finding and executing width-1 plans are indeed highly efficient in practice.

metaDecomp over DuckDB. Compared to DuckDB, we observe that metaDecomp also shows significant speedups, as demonstrated by Table 1, Fig. 10b, and Fig. 11b. This is again especially true for larger queries where DuckDB takes a long time, as shown by the points towards the right in Fig. 11b, due to the fact that DuckDB uses a simple greedy approach for such queries. Although the optimization is very fast and generates decent plans in many cases, it is also very likely to generate plans that are significantly worse than the optimal width-1 query plans.



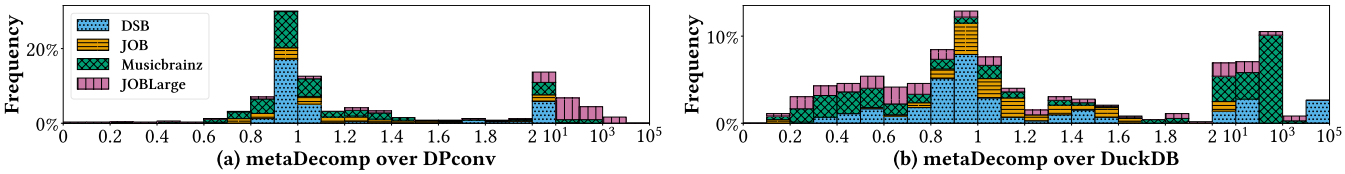
(a) Ratio between the costs[3]

(b) Speedup of execution[4]

Figure 9: Comparison of the optimal width-1 plan over the global optimal

	Benchmarks →	DSB	JOB	Musicbrainz	JOBLarge	Overall
Average[5] speedup of metaDecomp	Over DuckDB	2.75x	1.11x	6.39x	1.35x	2.79x
	Over DPconv	1.21x	1.18x	1.34x	26.21x	2.13x
Max speedup of metaDecomp	Over DuckDB	59559.26x	9.27x	2367.18x	1909.05x	59559.26x
	Over DPconv	3.57x	16.16x	180.52x	5511.57x	5511.57x
P95 latency	DuckDB	> 5 min	0.09 s	> 5 min	> 5 min	> 5 min
	DPconv	0.09 s	0.10 s	26.36 s	> 5 min	46.79 s
	metaDecomp	**0.04 s**	**0.08 s**	**2.65 s**	**2.31 s**	**1.79 s**
P99 latency	DuckDB	> 5 min	0.13 s	> 5 min	> 5 min	> 5 min
	DPconv	0.10 s	0.12 s	246.63 s	> 5 min	> 5 min
	metaDecomp	**0.04 s**	**0.11 s**	**13.75 s**	**23.84 s**	**7.09 s**
Total time	DuckDB	5723.25 s	4.99 s	22830.46 s	2484.86 s	31043.56 s
	DPconv	5.64 s	5.83 s	2090.03 s	6889.11 s	8990.61 s
	metaDecomp	**3.57 s**	**4.52 s**	**258.62 s**	**126.44 s**	**393.15 s**

Table 1: Key statistics of overall evaluation time (optimization plus execution)



(a) metaDecomp over DPconv

(b) metaDecomp over DuckDB

Figure 10: Histogram of the overall speedup of using metaDecomp over using DPconv and DuckDB

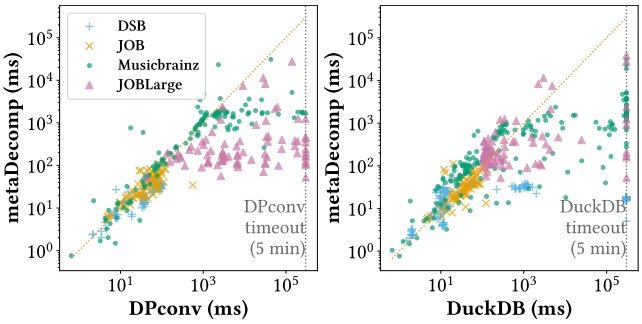


Figure 11: Total run time (optimization plus execution)

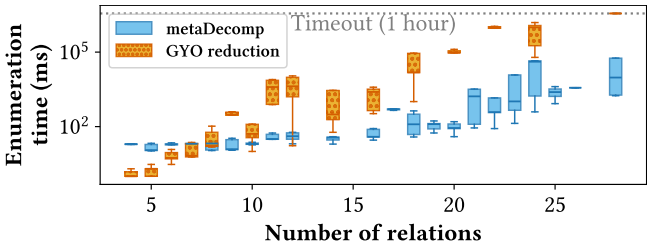


Figure 12: Box plot of the time to enumerate join trees

6.2.4 (RQ3) Meta-decompositions applied to other structure-based optimization techniques. Finally, we demonstrate ways in which meta-decompositions can help other recently-developed structure-based optimization techniques. A crucial step in those techniques is selecting a good join tree, as it can have a significant influence on the structures and hence the cost of the produced query plans. Thus, we first evaluate metaDecomp's efficiency in *enumerating join trees*, compared to the standard GYO-reduction-based approach from [12]. As shown in Fig. 12, metaDecomp can exhibit up to *two orders-of-magnitude of speedup* over the traditional approach. In particular, for queries involving 28 relations, the traditional approach already takes over 1 hour, whereas metaDecomp can enumerate the entire join order search space in between 2 seconds and 1 minute.

A more effective approach is to apply cost-based optimization directly on the meta-decomposition, as has been done in metaDecomp, since this eliminates the need to enumerate the exponential number of join trees. Therefore, in our second experiment, we use Yannakakis$^+$ [40] to rewrite queries, using the join tree selected by metaDecomp that induces the optimal width-1 query plan. *This modification achieves an average speedup of* 1.11x *on JOB queries.* These results demonstrate that structure-based evaluation algorithms, such as Yannakakis$^+$, can benefit significantly from our efficient cost-based optimization framework.

[3]The cost of the globally optimal plan divided by the cost of the optimal width-1 plan, analogous to speedups. A ratio of 1 indicates the two have the same cost, and is included in the bar between 0.9 and 1.0. Since width-1 plans are a restricted class, their costs are at least the global optimal and thus have ratios at most 1.

[4]Speedups over 2 are all included in the bar to the right of ratio 2.0.

[5]Geometric mean

7 Conclusions and Future Work

In this work, we introduce novel techniques for query optimization based on structural properties via meta-decompositions. They offer promising opportunities to integrate theoretically desirable structure-based optimization into practical database systems to efficiently evaluate large, complex queries. A natural next step is to extend our width notion and meta-decomposition-based framework for cyclic queries. While width-1 query plans are infeasible for cyclic queries based on Theorem 3.4, we may, e.g., identify minimal-width query plans and represent all minimal-width hypertree decompositions.

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A Missing Materials for Section 1

A.1 Number of Acyclic Queries in Major Benchmarks

Table 2 shows the numbers and proportions of acyclic queries in major benchmarks. One can clearly see that they indeed account for a majority.

Benchmark	# Queries	# Acyclic	% Acyclic
TPC-H [37]	22	21	95.45 %
JOB [27]	113	113	100 %
STATS-CEB [23]	2603	2603	100 %
Spider-NLP [46]	4712	4709	99.94 %
DSB [14] (SPJ queries only)	300	240	80 %
Musicbrainz [29]	375	243	64.8 %
Total	8125	7929	**97.59 %**

Table 2: Number of acyclic queries in major benchmarks

B Missing Proofs in Section 3

B.1 Proof of Theorem 3.1

Let $p^* \in V(\mathcal{P})$ be a node in the plan where the maximum width is achieved, i.e., $w(p^*) = w(\mathcal{P})$. By the definition of $w(p^*)$, there must exist a set of relations $S \subseteq E(H_{p^*})$ such that $|S| = w(p^*)$ and $I(p^*) \subseteq \bigcup_{e \in S} e$. Therefore, the result of $Q(p^*)$ must be a subset of the join of all relations in S. In the worst-case scenario, where all relations in S share no common attributes and their join is a Cartesian product, the size of this join is $O(N^{|S|})$, which gives an upper bound of $O(N^{w(\mathcal{P})})$.

B.2 Proof of Theorem 3.3

When $Q(p)$ only involves one relation, $Q(p) = q$ holds for all join trees because of the coverage condition of a join tree. Let T be the join tree satisfies the condition, for any $q \in V(T)$, the interface I_q between the induced hypergraph H_q for subtree T_q rooted by q and the residual hypergraph $\overline{H_q}$ satisfies $I_q \subseteq q$, because of the connectedness condition of the join tree. In addition, $Q(p) = Q(q)$ implies the induced hypergraph $H_p = H_q$, therefore, for every $p \in \mathcal{P}$, we can find a set $S = \{q\}$ that covers I_p, and $w(p) = 1$ holds for p, which completes the proof of if-direction.

For the only-if direction, given a width-1 query plan P, we can construct such a join tree T with induction. For any join operation $p \in V(\mathcal{P})$. Assuming there exists a valid join tree for both of the two child nodes c_1, c_2, denoted T_{c_1}, T_{c_2}. Let $q_1 \in V(T_{c_1})$ be the relation that covers the interface I_{c_1} and $q_2 \in V(T_{c_2})$ covers I_{c_2}, we make q_1 be the root of T_{c_1} and q_2 be the root of T_{c_2}, then $I_{c_1} \cap I_{c_2} = q_1 \cap q_2$, and we can merge T_1 and T_2 to obtain T' by connecting q_1 and q_2. In addition, there exists $q \in V(T_{c_1}) \cup V(T_{c_2})$ since $w(p) = 1$, such tree T' is a valid join tree for node p. Combining with the base case where T contains only one node, we then complete the proof.

B.3 Proof of Theorem 3.8

Following Theorem 3.3, we have

Lemma B.1. *For all width-1 query plans $\mathcal{P}$ induced by a join tree T, for all non-leaf nodes $t \in V(T)$ with distinct children c_1 and c_2, if $p_1, p_2 \in V(\mathcal{P})$ are the nodes on $\mathcal{P}$ such that $Q(p_1) = Q(c_1)$ and $Q(p_2) = Q(c_2)$, then there exists no $q \in \mathcal{P}$ such that $Q(q) = Q(p_1) \bowtie Q(p_2) = Q(c_1) \bowtie Q(c_2)$.*

Proof. Suppose there is such a q, then by Definition 3.2, there would have been an edge between c_1 and c_2 on T, contradicting with the assumption that c_1 and c_2 are distinct children of some node t. □

This means that, given a width-1 query plan $\mathcal{P}$ induced by a join tree T, for each $t \in V(T)$, with children $c_1, \ldots, c_{f(t)}$ and the corresponding $p \in V(\mathcal{P})$ such that $Q(t) = Q(p)$, if we consider the sub-query-plans $\mathcal{P}_{c_1}, \ldots, \mathcal{P}_{c_{f(t)}}$, induced by the sub-join-trees $T_{c_1}, \ldots, T_{c_{f(t)}}$ rooted at the children of t, each as a single relation, then the sub-query-plan $\mathcal{P}_p$ rooted at p is a *left-deep plan* consisting of $t, \mathcal{P}_{c_1}, \ldots, \mathcal{P}_{c_{f(t)}}$, and t is an operand of the deepest join operation.

This can be done by a dynamic programming algorithm, as shown in Algorithm 4. In the algorithm, $\mathsf{children}(v)$ is the set of all children of v.

In the algorithm, the outer loop (Line 1) is executed $|V(T)| = |Q|$ times. The complexity of the loop body is dominated by enumerating all possible T (Line 13) and $u \in T$ (Line 14). Noting that the number of children of v is bounded by the fan-out $f(p)$, the complexity of this part is given by

Algorithm 4: Finding the optimal width-1 query plan induced by a join tree of an acyclic query

input : A join tree T
output : The optimal width-1 query plan induced by T

```
foreach v ∈ V(T), in bottom-up order do
    foreach child c of v do
        if λ(c) ≠ ∅ then
            P(v, {c}) ← P(c, children(c) ∪ {c})
        else
            P(v, {c}) ← P(c, children(c))
    if λ(v) ≠ ∅ then
        P(v, {v}) ← query plan with one node v
    if λ(v) = ∅ then
        S ← 2^children(v) \ ∅
    else
        S ← {T ∪ {v} | T ∈ 2^children(v) \ ∅}
    foreach T ∈ S, ordered by increasing size do
        P(v, T) ← argmin_{p ∈ {P(v, T\{u}) ⋈ P(v, {u}) | u ∈ T}} cost(p)
return P(root(T), S(root(T)))
```

$$
\begin{aligned}
\sum_{i=1}^{f(v)} \binom{f(v)}{i} \times i &= \sum_{i=1}^{f(v)} \frac{f(v)!}{i!(f(v)-i)!} \times i \\
&= \sum_{i=1}^{f(v)} \frac{f(v)!}{(i-1)!(f(v)-i)!} \\
&= \sum_{i=1}^{f(v)} f(v) \times \frac{(f(v)-1)!}{(i-1)!((f(v)-1)-(i-1))!} \\
&= \sum_{i=1}^{f(v)} f(v) \times \binom{f(v)-1}{i-1} \\
&= f(v) \sum_{i=0}^{f(v)-1} \binom{f(v)-1}{i} \\
&= f(v) 2^{f(v)-1} \\
&\leq \max_{v \in V(T)} f(v) 2^{f(v)-1} = f(T) 2^{f(T)-1}.
\end{aligned}
$$

So, overall, the complexity is $O(f(T)2^{f(T)-1}|Q|)$.

C Missing Materials in Section 4

C.1 Missing Algorithms that Algorithm 2 is Based on

C.1.1 Enumeration of Non-Homomorphic Trees Based on Prüfer Sequences. For minor nodes p, as is illustrated by Example 2.3, all of its neighbours q for which $\kappa(q) = \chi(p)$ could be connected as an arbitrary tree with these neighbours being vertices. In order to enumerate all such possibilities without repetition, we (1) enumerate all non-homomorphic trees up to rerooting and (2) enumerate all rerootings for each tree enumerated in step (1). We start with the enumeration of non-homomorphic trees up to rerooting—namely, if two trees are simply rerootings of each other, they are treated as the same and only one of them will be enumerated. This can be done based on Prüfer sequences [32]. A Prüfer sequence is a sequence of $|V| - 2$ numbers, each of which is an integer between 1 and $|V|$, inclusive. It can be shown that each sequence bijectively encodes one tree (up to rerooting) of size $|V|$, with vertices labelled by integers ranging from 1 to $|V|$, inclusive. Therefore, we use the algorithm Algorithm 5 which simply enumerates all such numeric sequences and converts each sequence into a tree.

Algorithm 5: enumerateTrees(V): Enumerate all non-homomorphic trees, up to rerooting, with vertices V, using Prüfer sequences

input : Set of vertices V
output : A set $\mathcal{T}$ of all non-isomorphic trees with vertices V

```
Arbitrarily order the vertices in V and let v_i be the i-th vertex in V under this ordering
T ← ∅
foreach sequence P = p_1, ..., p_{|V|-2} of |V| - 2 integers, each with value in range [1, |V|] do
    foreach i = 1, ..., |V| do
        d(i) ← 1
    S ← [1, |V|]                    // maintains the set of indices i with d(v_i) = 1, in increasing order
    foreach i = 1, ..., |V| - 2 do
        d(s_i) ← d(p_i) + 1
        S ← S \ {i}
    Initialize a tree T with set of vertices V and no edge
    foreach i = 1, ..., |V| - 2 do
        j ← remove the least element in S
        d(i) ← d(i) - 1
        d(j) ← d(j) - 1
        if d(i) = 1 then
            Add i into S
        In T, add v_j as a child of v_i
    i, j ← the remaining two elements in S
    In T, add v_j as a child of v_i          // the direction does not matter
    T ← T ∪ {T}
return T
```

We then recursively enumerate all possible rerootings of a tree T such that the new root r' has $\kappa \subseteq \chi(r')$, where κ is the set of keys as required by the original algorithm Algorithm 2. This can be done as shown in Algorithm 6.

C.1.2 Enumeration of Rerootings. Algorithm 6 recursively enumerates all rerootings of a tree T such that the new root r' has $\kappa \subseteq \chi(r')$, where κ is the set of keys as required by Algorithm 2.

Algorithm 6: rerootings(T, p, κ): Enumerate all valid rerootings of a tree T such that the new root r' has $\kappa \subseteq \chi(r')$

input : A tree T, optionally the previous parent p of the root r of T before the previous rotation, the expected key κ
output : A set $\mathcal{T}$ of all valid rerootings of T such that the new root r' has $\kappa \subseteq \chi(r')$

```
T ← ∅
foreach child c of the root r of T do
    if κ ⊆ χ(c) and c ≠ p then
        T' ← T rerooted to c
        T ← T ∪ {T'} ∪ rerootings(T', r, κ)
return T
```

C.2 Proof of Lemma 4.6

We start with the following lemma:

Lemma C.1. *Given an acyclic hypergraph H, if there exist two ears $e_1, e_2 \in E(H)$ such that $o(e_1, H) = o(e_2, H)$, then $o(e_1, H) = o(e_2, H) = e_1 \cap e_2$.*

Proof. ($\subseteq$): $o(e_1, H) = e_1 \cap (\cup(E(H) \setminus \{e_1\})) \subseteq e_1$, and, similarly, $o(e_2, H) \subseteq e_2$. So it follows that $o(e_1, H) = o(e_2, H) \subseteq e_1 \cap e_2$.
($\supseteq$): Since $e_1 \in E(H) \setminus \{e_2\}$, we have $o(e_1, H) = o(e_2, H) = e_2 \cap (\cup(E(H) \setminus \{e_2\})) \supseteq e_1 \cap e_2$. □

A hypergraph is acyclic if and only if it has a join tree, satisfying conditions (H_1)–(H_3). Since e_1 and e_2 are ears in H such that $o(e_1, H) = o(e_2, H) \subseteq e_2$, in the GYO algorithm for H, e_1 can already be removed, with e_2 being its witness. Therefore, there exists a join tree T of H where there is a leaf node p_1 with $\lambda(p_1) = \{e_1\}$, and its parent p_2 has $\lambda(p_2) = \{e_2\}$. We construct a tree T' that is exactly the same as T except that p_1 and p_2 are merged into a single node m with $\lambda(m) = \{o(e_1, H)\}$ and $\chi(m) = o(e_1, H)$, and we show that T' is a valid join tree of H'. (H_1) and (H_3) are satisfied by construction. It remains to show that T' satisfies (H_2). Suppose by way of contradiction that this is not the case. By (H_2) on the join tree T, this means, on T', either (1) there exist $s, t \in V(T')$ such that m is on the path between s and t, and $\chi(s) \cap \chi(t) \nsubseteq \chi(m)$, or (2) there exists $s \in V(T')$ and t on the path between s and m, such that $\chi(s) \cap \chi(m) \nsubseteq \chi(t)$.

For case (1), since, by Lemma C.1, $\chi(m) = o(e_1, H) = e_1 \cap e_2 = \chi(p_1) \cap \chi(p_2)$, if $\chi(s) \cap \chi(t) \nsubseteq \chi(m)$, this means either $\chi(s) \cap \chi(t) \nsubseteq \chi(p_1)$ or $\chi(p) \cap \chi(q) \nsubseteq \chi(p_2)$. On T, since p_1 is a leaf node, p_2 should be on the path between s and t. By (H_2) for T, we have $\chi(s) \cap \chi(t) \subseteq \chi(p_2)$. So it can only be that $\chi(s) \cap \chi(t) \nsubseteq \chi(p_1)$, i.e., there exists some $v \in \chi(s) \cap \chi(t) \subseteq \chi(p_2)$ such that $v \notin \chi(p_1)$. But then, $v \in o(e_2, H)$ but $v \notin o(e_1, H)$, contradicting $o(e_1, H) = o(e_2, H)$.

For case (2), we first note that $\chi(s) \cap o(e_1, H) = \chi(s) \cap e_1$. This is because ($\subseteq$) $o(e_1, H) = e_1 \cap (\cup(E(H) \setminus \{e_1\})) \subseteq e_1$, and ($\supseteq$) $\chi(s) \cap e_1 \subseteq (\cup(E(H) \setminus \{e_1\})) \cap e_1 = o(e_1, H)$. Then, $\chi(s) \cap \chi(m) = \chi(s) \cap o(e_1, H) = \chi(s) \cap e_1 = \chi(s) \cap \chi(p_1)$. So, $\chi(s) \cap \chi(p_1) = \chi(s) \cap \chi(m) \nsubseteq \chi(p_2)$. Since p_1 is a leaf and p_2 is the parent of p_1 on T, p_2 must be on the path between s and p_1, so this contradicts (H_2) for T.

C.3 Proof of Theorem 4.7

We start by considering the while-loop (Line 3). At the start of each iteration, an ear must exist because the hypergraph H' must be acyclic, as guaranteed by Lemma 4.6. Furthermore, $E(H')$ always decreases in size by at least 1 in each iteration, because we only remove ears from $E(H')$, except when we have $e_1, e_2 \in S$ such that $o(e_1, H') = o(e_2, H')$, in which case we remove all (at least 2) those ears sharing the same intersection but add only one special hyperedge e_m to $E(H')$. Therefore, the body of the while loop is executed at most $|E(H)|$ times.

For each iteration of the while loop, all sets S (Line 4) can be enumerated in $O(|E(H')|)$, if we maintain two hash maps: (1) T_1, which maps each $o \subseteq V(H')$ to all hyperedges $e \in E(H')$ such that $o(e, H') = o$, and (2) T_2, which maps each $v \in V(H')$ to the number of hyperedges $e \in E(H')$ such that $v \in e$. Each time a hyperedge e is added or removed from $E(H')$, we update T_2 for each $v \in e$, and then update T_1 if some $v \in e$ is/was covered by exactly one hyperedge. These updates can be done in $O(|E(H')|)$, which is $O(|E(H)|)$.

With these observations, the runtime $O(|E(H)|^3)$ can be obtained by following the control flow.

C.4 Proof of Theorem 4.8

We show that M output by the algorithm satisfies (H_1)–(H_5).

(H_1). For each $e \in E(H)$, the algorithm has to remove e from $E(H')$ on Line 10, Line 15, or Line 21 before it terminates. In any of these cases, the algorithm will create a node p in M with $\lambda(p) = \{e\}$, on Line 9, Line 14, or Line 20.

Before proceeding with (H_2), we have the following lemmas:

Lemma C.2. *For each edge $(q, p) \in E(M)$, where $\chi(p) = \{e_p\}$ and $\chi(q) = \{e_q\}$, we have either*

(1) e_p is removed as ear by the algorithm before e_q is, or
(2) $\lambda(q) = \emptyset$ and $\kappa(q) = \chi(q)$

Additionally, $\kappa(p) = \chi(p) \cap \chi(q)$.

Proof. We first have $\kappa(p) \subseteq \chi(p)$ by definition on Lines 9, 14, 18, and 20. Since the algorithm adds the edge (q, p), by Lines 24 and 27, we have $\kappa(p) \subseteq \chi(q)$. Therefore, $\kappa(p) \subseteq \chi(p) \cap \chi(q)$.

We then consider the relative order in which e_p and e_q are removed from $E(H')$ in the algorithm.

(1) If e_p is removed before e_q is, then at the time e_p is removed, $e_q \in E(H')$, and therefore $\chi(p) \cap \chi(q) = e_p \cap e_q \subseteq e_p \cap (\cup(E(H') \setminus \{e_p\})) = \kappa(p)$. Since also $\kappa(p) \subseteq \chi(p) \cap \chi(q)$, we have $\kappa(p) = \chi(p) \cap \chi(q)$.

(2) If e_p is removed after e_q is, then, symmetric to case (1), we have $\chi(p) \cap \chi(q) \subseteq \kappa(q)$. Then, since we have shown that $\kappa(p) \subseteq \chi(p) \cap \chi(q)$, we have $\kappa(p) \subseteq \chi(p) \cap \chi(q) \subseteq \kappa(q)$. By Line 24 and Line 27, this is only possible if $\lambda(q) = \emptyset$ and $\kappa(p) = \chi(q)$. Since $\kappa(p) \subseteq \chi(p)$ and $\kappa(p) = \chi(q)$, it is true that $\kappa(p) = \chi(p) \cap \chi(q)$. □

It follows that

Corollary C.3. *p is a descendant of q on M if and only if e_p is removed as ear by the algorithm before e_q is.*

(H_2). The algorithm iteratively adds edges (q, p) to attach a connected subtree M_p rooted at p to a connected subtree M^q that contains q. Suppose by way of contradiction that (H_2) is violated at the end, and suppose it is when processing $p \in V(M)$, that the addition of edge (q, p) violates this condition for the first time. Namely, M_p and M^q each satisfies (H_2), but the resulting M violates (H_2), i.e., for some $s, t \in V(M)$, there exists u on the path betwween s and t such that $\chi(s) \cap \chi(t) \nsubseteq \chi(u)$. Without loss of generality, we take the shortest possible path from s to t where the condition is violated, where u is the node immediately before t on the path. Then, $\kappa(t) = \chi(t) \cap \chi(u)$.

Since M_p and M^q each satisfies (H_2), the only way it can be violated is if $s \in V(M_p)$, i.e., s is a descendent of p, $t \in M^q$, and thus p and q are on the path between s and t. For the purpose of the proof, we think of the subtree M_p as being added to the subtree M^q one node at a time, in a top-down order, starting from the root p. We then show that each time a node is added, the connectedness condition is preserved. Let $\lambda(p) = \{e_p\}$ and $\lambda(t) = \{e_t\}$.

As the base case, when p is added as a child of q, we consider two cases on the relative order in which e_p and e_t are removed from $E(H')$ by the algorithm.

(1) If e_p is removed as the ear by the algorithm before e_t is, then $\chi(p) \cap \chi(t) = e_p \cap e_t \subseteq \kappa(p) \subseteq \chi(q)$. And, for all u between q and t on the path between p and t, we have $\chi(p) \cap \chi(t) = \chi(p) \cap \chi(t) \cap \chi(t) \subseteq \chi(q) \cap \chi(t) \subseteq \chi(u)$, by the connectedness condition on M^q.

(2) If e_t is removed as the ear by the algorithm before e_p is, then $\chi(p) \cap \chi(t) = e_p \cap e_t \subseteq \kappa(t)$. By Corollary C.3, t is a descendant of q, and so the parent u of t is on the path between p and t. Then, $\chi(p) \cap \chi(t) \subseteq \kappa(t) = \chi(t) \cap \chi(u) \subseteq \chi(u)$

As the induction step, assuming that the condition holds for the connected subtree with nodes in M^q and all p' that are added before s. Following a similar argument as for the base case, substituting q with the parent of s, we can also derive that $\chi(s) \cap \chi(t) \subseteq \chi(u)$ for all u.

(H_3'). This is guaranteed by construction on Lines 9, 14, 18, and 20.

(H_4)(a). By Lemma C.2, for each $p \in V(M)$ and its parent q, we have $\kappa(p) = \chi(p) \cap \chi(q)$. ($\subseteq$): $\kappa(p) = \chi(p) \cap \chi(q) \subseteq \chi(q)$. ($\supseteq$): For all $s \in V(M) \setminus V(M_p)$, since q is the parent of p, q has to be on the path between p and s. Then, by the connectedness condition (H_2) on M, we have $\chi(p) \cap \chi(s) \subseteq \chi(q)$. Therefore, $\kappa(p) = \chi(p) \cap \chi(q) \supseteq \chi(p) \cap \left(\bigcup_{s \in V(M) \setminus V(M_p)} \chi(s)\right) = \chi(p) \cap \chi(V(M) \setminus V(M_p))$.

(H_4)(b). This is guaranteed by Lines 24 and 27.

(H_5). Lines 4–11 and Lines 17–18 create such minor nodes when necessary. Lines 6–7 and the condition on Line 17 before creating a minor node on Line 18 ensures there are no duplicate minor nodes.

C.5 Proof of Theorem 4.9

Vertices $V(M)$ in M are either physical nodes or minor nodes. Let $V^P(M)$ be the set of physical nodes and $V^M(M)$ be the set of minor nodes, such that $V(M) = V^P(M) \cup V^M(M)$. Each time we create a physical node p with $\lambda(p) = \{e\}$ (Lines 9, 14, and 20), we remove e from $E(H')$ (Lines 10, 15, and 21). So there are at most $|E(H)|$ physical nodes, $|V^P(M)| \leq |E(H)|$. Now let us consider the number of minor nodes, $|V^M(M)|$. Note that each minor node $m \in V^M(M)$ must have at least two children (see Line 4 and Line 17), i.e., the fanout $f(m) \geq 2$. Then, the total number of vertices

$$\begin{aligned} |V(M)| &= |V^P(M)| + |V^M(M)| \\ &= |E(M)| + 1 = 1 + \sum_{p \in V(M)} f(p) \\ &> 1 + \sum_{m \in V^M(M)} f(m) \geq 1 + \sum_{m \in V^M(M)} 2 = 2|V^M(M)| + 1. \end{aligned}$$

Then, $|V^P(M)| + |V^M(M)| > 2|V^M(M)| + 1$, so $|V^M(M)| < |V^P(M)| - 1 = |E(H)| - 1$. So, $|V(M)| = |V^P(M)| + |V^M(M)| < |E(H)| + |E(H)| - 1 = 2|E(H)| - 1$. And $|E(M)| = |V(M)| - 1 \leq 2|E(H)| - 2$.

C.6 Proof of Theorem 4.11

We prove by showing that all trees T output by Algorithm 2 satisfy (H_1), (H_2), and (H_3).

For (H_1), note that this condition must already hold for the meta-decomposition M. In Algorithm 2, all physical nodes, for which $\lambda(r) \neq \emptyset$, will be kept in $\mathcal{T}$, on Line 23. Minor nodes, for which $\lambda(r) = \emptyset$, will no longer exist in the output. If $\kappa(r) \neq \chi(r)$, trees in $\mathcal{T}_p$, and thus $\mathcal{T}$, will not contain minor nodes (Line 10) but only nodes in the subtrees rooted at the children of r (Lines 11–20). If $\kappa(r) = \chi(r)$, the minor node r on Line 8 will eventually be removed and replaced by v on Line 33. Therefore, the trees output by Algorithm 2 contain only physical nodes. (H_3) then follows, by (H_3') for M.

It now remains to show (H_2), the connectedness condition. We prove by structural induction on vertices $v \in V(M)$, i.e., for each $v \in V(M)$, all trees T returned by the function enumRec satisfy (H_2). In the base case, for all leaf nodes v, v has to be physical, and the function simply returns a tree with one single node, so these conditions hold. For non-leaf nodes v, we assume as induction hypothesis that for all children c of v, all trees returned by the function call enumRec(c) satisfy (H_2). And we now show that it remains satisfied throughout the operations in enumRec(v).

We start with the trees in $\mathcal{T}$ constructed on Lines 3–23. If $\lambda(v) \neq \emptyset$, this vacuously holds as there is only one tree with one vertex r in $\mathcal{T}$ (Line 23). If $\lambda(v) = \emptyset$, we will show that it still holds for all T'_c (constructed on Lines 11–20), assuming that it holds for $T_c \in \mathcal{T}_c$ from the previous iteration. It suffices to show that, for each (T_P, c, T_c, d, T_d, u) processed in the loop, for all $p \in T_c$ and $q \in T_d$, all s on the path between p and q has $\chi(p) \cap \chi(q) \subseteq \chi(s)$. Let the root of T_c be r_c and the root of T_d be r_d. The path from p to q needs to go through u then r_d. For s on the path between p and u, we have $\chi(p) \cap \chi(q) \subseteq \chi(v) = \kappa(d) \subseteq \chi(u)$. Since T_c satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(u) \cap \chi(q) \subseteq \chi(s)$. For s on the path between r_d and q, we have $\chi(p) \cap \chi(q) \subseteq \chi(v) = \kappa(d) \subseteq \chi(r_d)$. Since T_d satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(r_d) \cap \chi(q) \subseteq \chi(s)$.

We now continue to show that lines Line 25–Line 40 preserve the condition (H_2) for all $T \in \mathcal{T}$, by showing that the condition is preserved for all $T' \in \mathcal{T}'$ after each child $c \in X$ is processed, assuming that all the previous $T \in \mathcal{T}$ satisfy (H_2).

Let the root of T_c be r_c. By the connectedness condition on M, we have $\chi(p) \cap \chi(q) \subseteq \chi(v)$, and $\chi(p) \cap \chi(q) \subseteq \chi(c)$. Since c is a child of v, $\chi(p) \cap \chi(q) \subseteq \chi(v) \cap \chi(c) = \kappa(c)$.

(1) If $\lambda(c) \neq \emptyset$ or $\kappa(c) \neq \chi(c)$, T_c is added directly as a child of u on T' (Line 38). For all $p \in V(T')$ and $q \in V(T_c)$, the path from p to q needs to go through u then r_c. For s on the path between p and u, we have $\chi(p) \cap \chi(q) \subseteq \kappa(c) \subseteq \chi(u)$. Since T' satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(u) \cap \chi(q) \subseteq \chi(s)$. For s on the path between r_c and q, we have $\chi(p) \cap \chi(q) \subseteq \kappa(c) \subseteq \chi(r_c)$. Since T_c satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(r_c) \cap \chi(q) \subseteq \chi(s)$.

(2) If $\lambda(c) = \emptyset$ and $\kappa(c) \neq \chi(c)$, for each child d of the root of T_c, we attach T_d as a child of u. Let the root of T_d be r_d. Then, $\kappa(d) \subseteq \chi(r_d)$ and $\kappa(d) \subseteq \chi(u)$. For all $p \in V(T')$ and $q \in V(T_d)$, the path from p to q needs to go through u then r_d. For s on the path between p and u, we have $\chi(p) \cap \chi(q) \subseteq \kappa(c) = \kappa(d) \subseteq \chi(u)$. Since T' satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(u) \cap \chi(q) \subseteq \chi(s)$. For s on the path between r_c and q, we have $\chi(p) \cap \chi(q) \subseteq \kappa(c) = \kappa(d) \subseteq \chi(r_d)$. Since T_c satisfies (H_2), we have $\chi(p) \cap \chi(q) \subseteq (\chi(p) \cap \chi(q)) \cap \chi(q) \subseteq \chi(r_d) \cap \chi(q) \subseteq \chi(s)$.

C.7 Proof of Theorem 4.12

We start from a few lemmas:

Lemma C.4. *For any non-root node p on a meta-decomposition M with parent q, $\kappa(p) = \chi(p) \cap \chi(q)$.*

Proof. ($\supseteq$): Since $q \in V(M) \setminus V(M_p)$, $\chi(p) \cap \chi(q) \subseteq \chi(p) \cap \chi(V(M) \setminus V(M_p)) = \kappa(p)$. ($\subseteq$): For any $v \in \kappa(p) = \chi(p) \cap \chi(V(M) \setminus V(M_p))$, i.e., any $v \in \chi(p) \cap \chi(s)$ for some $s \in V(M) \setminus V(M_p)$, the parent q of p lies on the path on M between p and s. Then, by the connectedness condition (H_2) for M, $v \in \chi(q)$. □

Lemma C.5. *For all $p \in V(M)$ with $\lambda(p) = \emptyset$ and $\kappa(p) = \chi(p)$, all children c of p satisfy $\kappa(c) = \chi(p)$.*

Proof. Suppose by way of contradiction that there exists such a $p \in V(T)$ with child c such that $\kappa(c) \neq \chi(p)$. By Lemma C.4, we have $\kappa(c) = \chi(c) \cap \chi(p) \subseteq \chi(p)$. Since $\kappa(p) = \chi(p) \neq \emptyset$, p is not the root of M and thus has a parent q. Again by Lemma C.4, we have $\kappa(p) = \chi(p) \cap \chi(q) \subseteq \chi(q)$. But then, $\kappa(c) \subseteq \chi(p) = \kappa(p) \subseteq \chi(q)$, contradicting ($H_4$)(b) since $\kappa(c) \neq \chi(p)$ but $q \in V(T) \setminus V(T_p)$. □

Lemma C.6. *Given a meta-decomposition M, if some non-root node $p \in V(M)$ has $\lambda(p) = \emptyset$ and $\kappa(p) = \chi(p)$, then its parent q has either $\lambda(q) \neq \emptyset$ or $\kappa(q) \neq \chi(q)$.*

Proof. Suppose by way of contradiction that, on a meta-decomposition M, there is $p \in V(M)$ such that $\lambda(p) = \emptyset$ and $\kappa(p) = \chi(p)$, and its parent q has $\lambda(q) = \emptyset$ and $\kappa(q) = \chi(q)$. Then, $\kappa(p) \subseteq \chi(q) = \kappa(q)$. Since $\kappa(q) = \chi(q) \neq \emptyset$, q has a parent as well. Let the parent of q be s. Then, $\kappa(q) = \chi(q) \cap \chi(s) \subseteq \chi(s)$. So, $\kappa(p) \subseteq \kappa(q) \subseteq \chi(s)$. By ($H_5$), $\kappa(p) \neq \kappa(q) = \chi(q)$, contradicting ($H_4$)(b). □

Lemma C.7. *Given a meta-decomposition M of an acyclic hypergraph H, for all $v \in V(M)$,*

(1) if $\lambda(v) = \emptyset$ and $\kappa(v) = \chi(v)$, then, for each join tree T of H, there exists some $S = \{p \in V(M) \mid \lambda(p) \neq \emptyset, \chi(v) \subseteq \chi(p)\}$ such that the set $V_v^P = \{p \in V(M_v) \mid \lambda(p) \neq \emptyset\} \cup S$ induces a connected subtree T_v of T;

(2) if $\lambda(v) \neq \emptyset$ or $\kappa(v) \neq \chi(v)$, then, for each join tree T of H, the set $V_v^P = \{p \in V(M_v) \mid \lambda(p) \neq \emptyset\}$, i.e., physical nodes on the subtree M_v rooted at v, induces a connected subtree T_v of T.

Proof. We prove by structural induction on M. In the base case, for leaf nodes $v \in V(M)$, there is only one node on the subtree M_v, so the statement is vacuously true. For non-leaf nodes $v \in V(M)$, we assume as induction hypothesis that the statement is true for all children c of v. Let c and d be the children of v such that $p \in V(M_c)$ and $q \in V(M_d)$. We now consider two cases of v.

(1) If $\lambda(v) = \emptyset$ and $\kappa(v) = \chi(v)$, then, for any s on the path between p and q on T such that $s \notin M_v$, we have $\chi(p) \cap \chi(q) \subseteq \chi(s)$ by the connectedness condition on T. By (the contrapositive of) Lemma C.6, $\lambda(c) \neq \emptyset$ or $\kappa(c) \neq \chi(c)$, so by the inductive hypothesis, V_c^P induces a connected subtree T_c of T. Similarly, V_d^P induces a connected subtree T_d of T. If $c = d$, then p and q are in the same subtree rooted at the same child of v, and the statement would hold. If $c \neq d$, on the path from p to q on T, let the last node on T_c be x and the first node on T_d be y. s is between x and y. Then, the path from c and d has to go through x and y, and hence also s. Therefore, $\chi(c) \cap \chi(d) \subseteq \chi(s)$. By Lemma C.5, in this case, $\kappa(v) = \kappa(c) = \kappa(d) = \chi(c) \cap \chi(d) \subseteq \chi(s)$.

(2) If $\lambda(v) \neq \emptyset$ or $\kappa(v) \neq \chi(v)$, we would like to show that for any s on the path between p and q on T, $s \in V(M_v)$. Suppose by way of contradiction that this is not the case, i.e., there exists some s on the path between p and q on T, but $s \in V(M) \setminus V(M_v)$. By

the connectedness condition on T, we have $\chi(p) \cap \chi(q) \subseteq \chi(s)$. Furthermore, on M, $\chi(p) \cap \chi(s) \subseteq \chi(v)$ and $\chi(q) \cap \chi(s) \subseteq \chi(v)$, so $\chi(p) \cap \chi(q) = \chi(p) \cap \chi(q) \cap \chi(s) \subseteq \chi(v)$. Let the parent of p on M be t. Then, $\chi(t) \cap \chi(s) \subseteq \chi(v)$. Then, $\kappa(p) = \kappa(p) \cap \chi(s) = \chi(p) \cap \chi(t) \cap \chi(s) = (\chi(p) \cap \chi(s)) \cap (\chi(t) \cap \chi(s)) \subseteq \chi(v) \cap \chi(v) = \chi(v)$. But, by ($H_4$), this means either $t = v$ or t is a minor node with $\kappa(t) = \chi(t)$. In either case, we have $\chi(t) \cap \chi(q) \subseteq \chi(v)$, because (i) if $t = v$, $\chi(t) \cap \chi(q) = \chi(v) \cap \chi(q) \subseteq \chi(v)$, and (ii) if t is a minor node with $\kappa(t) = \chi(t)$, then $\chi(t) \cap \chi(q) = \kappa(t) \cap \chi(q) = \kappa(p) \cap \chi(q) \subseteq \chi(p) \cap \chi(q) \subseteq \chi(v)$. Also, we have $\kappa(t) = \chi(p) \cap \chi(t) \subseteq \chi(q)$, because either t is on the path between p and q, or p is on the path between t and q. Let the parent of v be u. Then, we have $\kappa(t) = \kappa(t) \cap \chi(q) = \chi(t) \cap \chi(q) = (\chi(t) \cap \chi(q)) \cap (\kappa(p) \cap \chi(q)) \subseteq (\chi(t) \cap \chi(q)) \cap (\chi(p) \cap \chi(q)) \subseteq \chi(v) \cap \chi(s) \subseteq \chi(u)$. But, since it cannot be that $\kappa(v) = \chi(v)$ in this case, this violates (H_4), contradiction. □

Lemma C.8. *Given a meta-decomposition M of hypergraph H, for all $v \in V(M)$, let $V_v^P = \{p \in V(M_v) \mid \lambda(p) \neq \emptyset\}$. Then,*

(1) if $\lambda(v) = \emptyset$ and $\kappa(v) = \chi(v)$, then Algorithm 2 enumerates all join trees of the hypergraph H_v with $E(H_v) = \lambda(V_v^P) \cup \{\chi(v)\}$ and $V(H_v) = \cup E(H_v)$;

(2) if $\lambda(v) \neq \emptyset$ or $\kappa(v) \neq \chi(v)$, then Algorithm 2 enumerates all join trees of the hypergrpah H_v with $E(H_v) = \lambda(V_v^P)$ and $V(H_v) = \cup E(H_v)$.

Proof. We will show that each join tree T of H can be enumerated by the algorithm based on the meta-decomposition M of H. We prove by structural induction on vertices $v \in V(M)$. If v is a leaf node, there is only one node in the subtree M_v, so this statement is vacuously true. If v is a non-leaf node, consider the following cases.

(1) If $\lambda(v) = \emptyset$ and $\kappa(v) = \chi(v)$, by Lemma C.6, all children c of v have either $\lambda(c) \neq \emptyset$ or $\kappa(c) \neq \chi(c)$, so, by the induction hypothesis, Algorithm 2 enumerates all join trees of hypergraph H_c with $E(H_c) = \bigcup_{p \in V_c^P} \lambda(p)$ and $V(H_c) = \bigcup_{e \in E(H_c)} e$. Also, note that it is possible to obtain a meta-decomposition M'_v of the hypergraph H_v from M_v by simply replacing v by a physical node v' with $\lambda(v') = \{\chi(v)\}$, $\chi(v') = \chi(v)$, and $\kappa(v') = \emptyset$. By Lemma C.7, for each child c of v, on any join tree T_c of hypergraph H_c, V_c^P induces a connected subtree T_c of T. Furthermore, T_c is a join tree of the meta-decomposition M_c, and can hence be enumerated on Line 6. So it remains to show that the algorithm enumerates all possible ways to combine these subtrees along with v. For each possible join tree T_v of H_v, without loss of generality, assume that T_v is rooted at v'. T_v can be constructed from the algorithm as follows. We first construct a tree T_P, by a top-down traversal of T_v, where, for each node p, if (i) $\chi(v) \subseteq \chi(p)$, (ii) $p \in M_c$, i.e., in the subtree M_c rooted at some child c of v, and (iii) its parent q is in a different subtree M_d rooted at some child $d \neq c$ of v, then we add an edge from d to c. If $p = v$, and its parent $q \in V(M_d)$ for some child d of v, we add an edge from d to v. If $p \in V(M_c)$ for some child c of v, and its parent $q = v$, we add an edge from v to c. Note that, for all pairs of c and d where we add edges, $\kappa(c) = \kappa(d) = \chi(v)$. Now, this tree T_P is a spanning tree of the clique with vertices $C \cup \{v\}$, where C is the set of all children of v, so it can be enumerated by the Prüfer sequence on Line 8. For each child c of v, by Lemma C.7, V_c^P induces a connected subtree T_c of T_v. We traverse the vertices $c \in V(T_P)$ in bottom-up order. For each c and each child d of c on T_P, let r_d be the root of the connected subtrees T_d of T_v induced by V_d^P, and u be the parent of r_d on T_v. The algorithm will be able to reroot T_d to r_d, find u (since $\kappa(d) = \kappa(v) = \chi(v) = \chi(c) \subseteq \chi(u)$), and attach T_d as a child of u. Finally, by Lemma C.5, in this case, $S = C$ and $X = \emptyset$, so the second part will not be executed at all.

(2) If $\lambda(v) = \emptyset$ and $\kappa(v) \neq \chi(v)$, starting from Lines 11–20, note that it is not possible to have a child $c \in C$ such that $\kappa(c) = \chi(c)$, because if that is the case, $\kappa(c) = \chi(c) = \chi(v)$, violating the uniqueness condition (H_5). Therefore, for all $c \in C$, we have $\lambda(c) \neq \emptyset$ or $\lambda(c) \neq \chi(c)$. Furthermore, we claim that the vertices $V_C^P = \bigcup_{c \in C} V_c^P$ induces a connected subtree T_C on any join tree T_v of M_v. This is because, if this is not the case, i.e., if there is some s on the path between some $p \in V_c^P$ and $q \in V_d^P$, where c and d are children of v with $\kappa(c) = \kappa(d) = \chi(v)$, such that $\kappa(s) \subsetneq \chi(v)$, then $\chi(p) \cap \chi(q) = \chi(v) \not\subseteq \kappa(s)$, violating the connectedness condition (H_2). Therefore, we can follow the same method as in case (1) to construct T_P with vertices C and show that, at the end of this part, it is possible to obtain a partial join tree T with vertices in V_C^P.

Now consider all children $c \in X = \{c \in C \mid \kappa(c) \subsetneq \chi(v)\}$.

(i) If $\lambda(c) = \emptyset$ and $\kappa(c) = \chi(c)$, by Lemma C.6, all children d of c has either $\lambda(d) \neq \emptyset$ or $\kappa(d) \neq \chi(d)$, so, by Lemma C.7, V_d^P induces a connected subtree in any join tree. We build a tree T_c by (a) collecting all maximal connected subtrees of the join tree with nodes $p \in V_c^P$ and (b) adding a common parent c as the root node. This is a valid join tree of the hypergraph H_c with $E(H_c) = \left(\bigcup_{p \in V_c^P} \lambda(p)\right) \cup \{\chi(c)\}$ and $V(H_c) = \bigcup_{e \in E(H_c)} e$, and, by our inductive hypothesis, can be enumerated. For each child d of c on T_c, let the parent of d on T be u. The algorithm will be able to find such u to attach T_d, because $\kappa(d) = \chi(d) \cap \chi(c) \subseteq \chi(u)$, by the connectedness condition on T.

(ii) If $\lambda(c) \neq \emptyset$ or $\kappa(c) \neq \chi(c)$, similarly to case (i), by Lemma C.7, V_c^P induces a connected subtree on T, which, by our inductive hypothesis, can be enumerated. Then, the algorithm will be able to find such u to attach T_c, because $\kappa(c) = \chi(c) \cap \chi(v) \subseteq \chi(u)$, by the connectedness condition on T.

(3) If $\lambda(v) \neq \emptyset$, we can follow a similar argument for the children $c \in X$ in case (2). □

Finally, based on Lemma C.8, taking $v = r$, the root of M (for which $\kappa(r) = \emptyset \neq \chi(r)$), we finish the proof of completeness for the algorithm.

C.8 Polynomial-Delay Algorithm to Enumerate All Join Trees Based on a Meta-Decomposition

Algorithm 7 shows an alternative implementation of Algorithm 2, which does not require all possible join trees induced by each children to have been enumerated before being able to proceed to construct join trees induced by the current node. Instead, Algorithm 7 uses callback functions (higher-order functions) as continuation after each join tree induced by a child is enumerated.

Algorithm 7: Enumerating all join trees based on a meta-decomposition, in polynomial delay

```
input  : A meta-decomposition M = (V(M), r, E(M), λ, χ, κ)
output : All possible non-homomorphic join trees up to rerooting
Function enumRec(v: a node on M, callback(T) : a callback function once enumRec obtains a tree T):
    C ← set of children c of v on M, sorted by partial order ⊇ on κ(c)
    if λ(v) = ∅ then
        S ← {c ∈ C | κ(c) = χ(v)}  // they can be found at the head of the sorted C
        X ← C \ S
        if κ(v) = χ(v) then
            T_v ← tree with one vertex v
            enumSpanningTrees(S ∪ {v}, X, callback)
        else
            enumSpanningTrees(S, X, callback)
    else
        T_v ← tree with one vertex v
        attachChildren(T_v, C, callback)
Function enumSpanningTrees(S: set of nodes, C: set of children, callback):
    foreach tree T_p with vertices V given by a Prüfer sequence do
        enumMinorOriginSubtrees(T_p, S, X, callback)
Function enumMinorOriginSubtrees(T: partial tree, S: remaining origins, X: remaining children, callback):
    if S = ∅ then
        attachChildren(T, X, callback)
    else
        c ← some node in S
        S' ← S \ {c}
        enumRec(c, T_c ⇒ enumOrigins(T with c replaced by T_c, S', X, callback))
Function attachChildren(T: partial tree, X: remaining children, callback):
    if X = ∅ then
        callback(T)  // We finish constructing a valid join tree. Now we call the callback function.
    else
        c ← some node in S
        S' ← S \ {c}
        if λ(c) = ∅ and κ(c) = χ(c) then
            enumRec(c, T_c ⇒ { foreach u ∈ V(T) such that κ(c) ⊆ χ(u) do attachChildren(T with u replaced by T_c, S', callback)}
        else
            foreach u ∈ V(T) such that κ(c) ⊆ χ(u) do
                enumRec(c, T_c ⇒ { foreach T' ∈ rerootings(T, κ(c), ∅) do attachChildren(T with c replaced by T', S', callback}
enumRec(r, T ⇒ { foreach T' ∈ rerootings(T, ∅, ∅), emit T'}
```

D Missing Materials for Section 5

D.1 Supporting Re-branching for Algorithm 3

For each node p, if there exists a subtree T_q that can be attached as a child of p, we simply use an additional bit in the DP table (plan) to indicate whether T_q is included. And we call optimizeLocal with or without the relations in T_q accordingly. We note again that this is a very rare possibility in practical queries.

D.2 Greedy Algorithm for Local Join Ordering

The implementation of local join ordering optimization using a greedy algorithm is as shown in Algorithm 8.

Algorithm 8: Greedy implementation of optimizeLocal

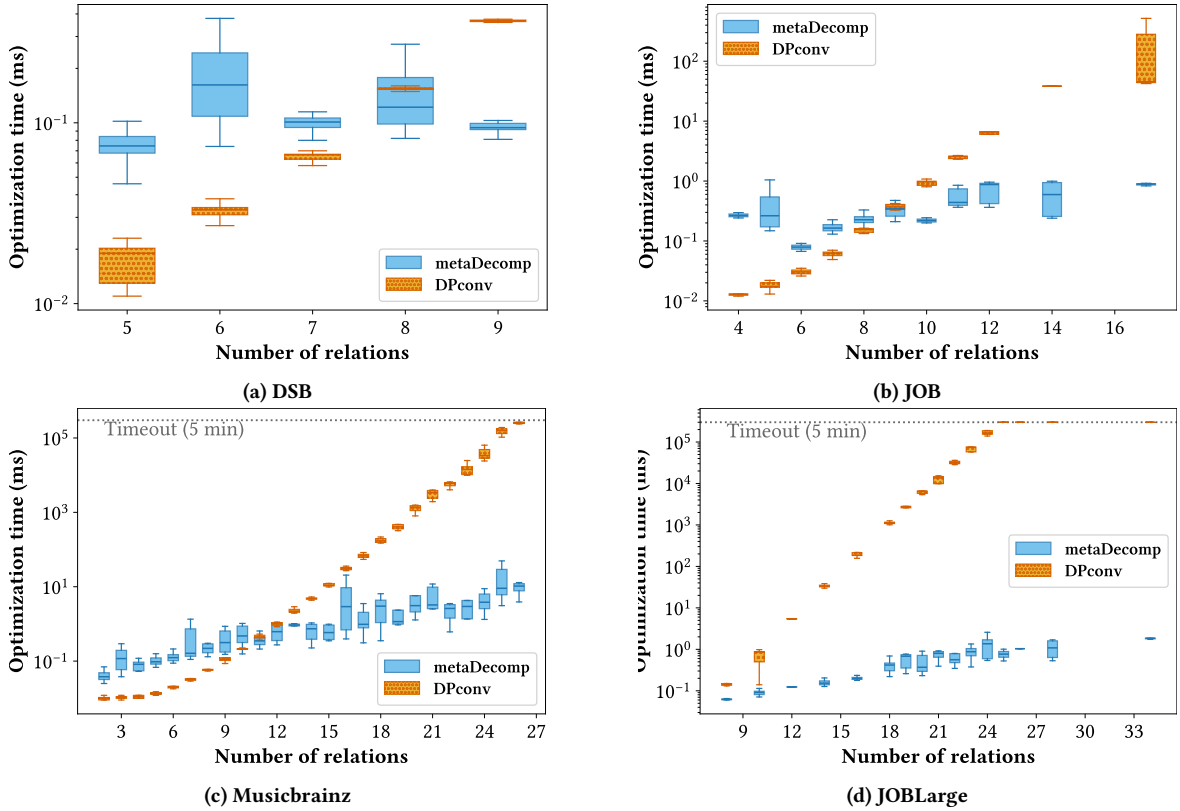
input : Node q on meta representation, a set of query plans $\mathcal{P}$
output : A query plan that joins the relation in q with all query plans in $\mathcal{P}$

```
if q is a minor node then                    // λ(q) = ∅, there is no relation in q
    P ← remove P_1 from 𝒫 with the minimum result cardinality
else
    P ← the relation in λ(q)
while 𝒫 ≠ ∅ do
    P_i ← remove P_i from 𝒫 that minimizes the cost to join P with P_i
    P ← the plan P_i ⋈ P
return P
```

E Missing Materials for Section 6

E.1 Optimization Time of metaDecomp and DPconv for Different Query Sizes on Each Benchmark

Fig. 13 shows the detailed breakdown of optimization time measured for queries in each of the four benchmarks used in the paper. The trends that can be observed in each of the four benchmarks and the overall figure (Fig. 8) are mostly consistent, with the running time of DPconv starting to overtake that of metaDecomp starting from approximately 8 relations, and DPconv starting to timeout (taking more than 5 minutes) from approximately 25 relations.

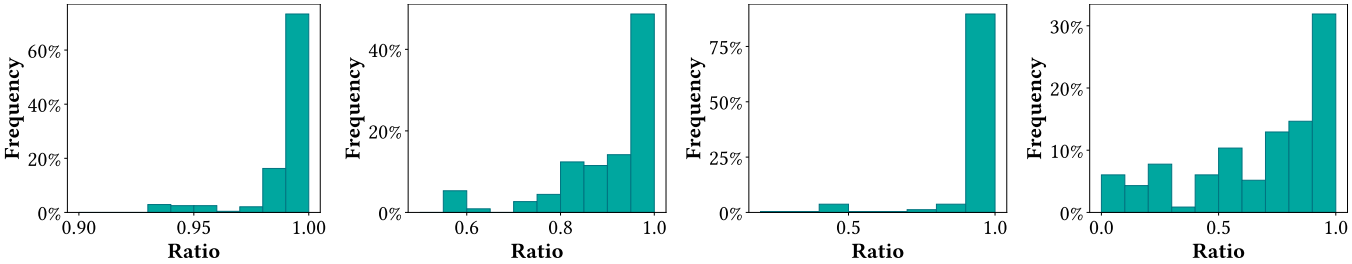


(a) DSB (b) JOB (c) Musicbrainz (d) JOBLarge

Figure 13: Box plots of the optimization time of metaDecomp and DPconv for queries in each benchmark

E.2 Comparison of the Globally Optimal Query Plans and the Optimal Width-1 Plans

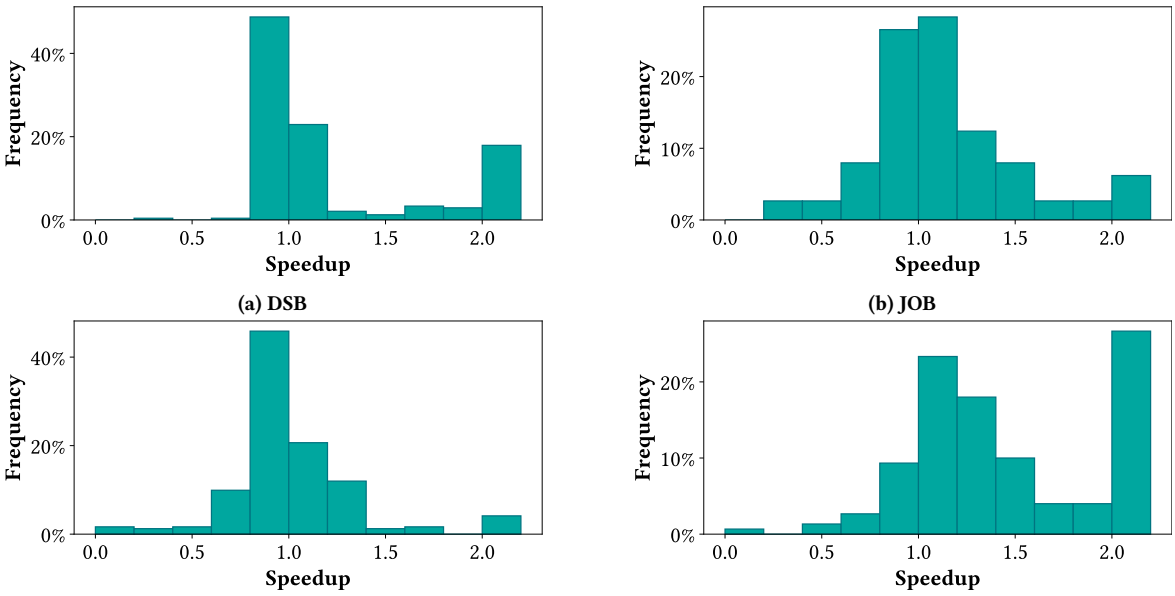
E.2.1 Cost Values. Fig. 14 shows histograms of the ratios between the costs of the globally optimal query plans over the costs of the optimal width-1 plans for queries in each benchmark. This measure is analogous to speedups. A ratio of 1.0 indicates that the optimal width-1 plan has the same cost as the globally optimal plan, and such cases are included in the bar to the left of 1. This measure is analogous to speedups, although the ratio should be at most 1 as the width-1 query plans are a restricted class. However, we can still observe that the optimal width-1 query plans tend to have costs very close to the global optimal, especially in the DSB and Musicbrainz benchmarks.



(a) DSB (b) JOB (c) Musicbrainz (d) JOBLarge

Figure 14: Ratios between the costs of the globally optimal query plans over the costs of the optimal width-1 plans for queries in each benchmark.

E.2.2 Execution Time. Fig. 15 shows histograms of the speedups of the actual execution of the optimal width-1 query plans over that of the globally optimal plans. In most cases, we observe speedups close to 1, which indicates that the actual performance between the two indeed tends to be similar. Furthermore, although one should not expect speedups as the width-1 plans are being compared with the global optimal, we actually observe significant speedups, even many over 2, especially in the JOBLarge benchmark. This is thanks to the possibilities of early projection of irrelevant attributes made possible by enforcing the connectedness conditions in width-1 query plans.



(a) DSB (b) JOB (c) Musicbrainz (d) JOBLarge

Figure 15: Speedups of execution time of the optimal width-1 plans over the globally optimal query plans for queries in each benchmark. Speedups over 2 are all included in the bar to the right of ratio 2.0.

E.3 Total Query Evaluation Time Using metaDecomp versus Using DPconv on Each Benchmark

Fig. 16 shows histograms of the speedups of the overall query execution time using metaDecomp over using DPconv for queries in each of the four benchmarks. In DSB, JOB, and Musicbrainz, most of the queries show a speedup close to 1. In JOBLarge, however, we observe a large number of significant speedups of multiple orders of magnitudes, mostly due to the high optimization time of DPconv for the large query sizes in this benchmark.

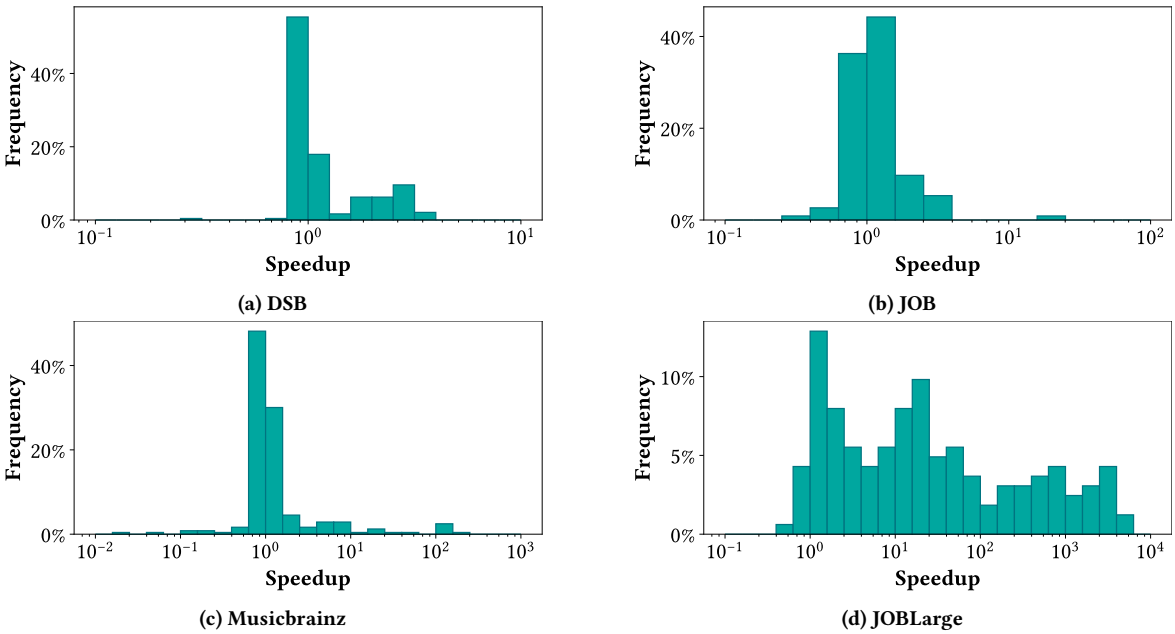
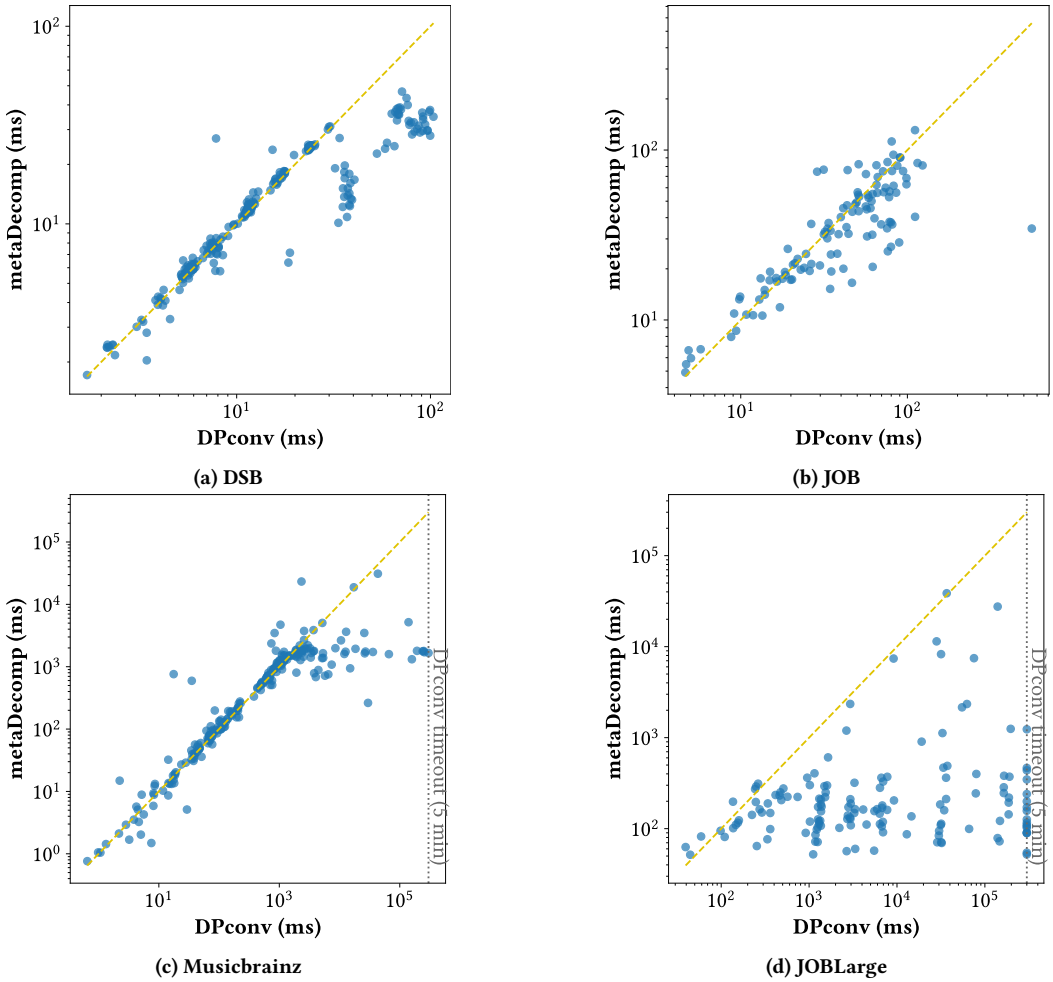


Figure 16: Histograms of the speedups of overall query evaluation time of metaDecomp over DPconv on each benchmark

Fig. 17 shows a scatter plot of the overall query evaluation time using metaDecomp and using DPconv measured for queries in each of the four benchmarks. The trends that can be observed in each of the four benchmarks and the overall figure (Section 6.2.3) are mostly consistent. The two approach have similar performance for queries that takes DPconv less than 1 second (10^3 ms). For those taking DPconv more than 1 second, metaDecomp starts to show significant speedups. Notably, for complex queries in Musicbrainz (Fig. 17c) and JOBLarge (Fig. 17d) for which DPconv times out (taking more than 5 minutes), metaDecomp can evaluate them within merely 1 second.



(a) DSB (b) JOB (c) Musicbrainz (d) JOBLarge

Figure 17: Scatter plot of overall query evaluation time of using metaDecomp versus using DPconv.

E.4 Total Query Evaluation Time Using metaDecomp versus Using DuckDB on Each Benchmark

Fig. 18 shows histograms of the speedups of the overall query execution time using metaDecomp over using DuckDB for queries in each of the four benchmarks. Most of the queries show a speedup close to 1, with some notable speedups especially for larger queries in the Musicbrainz benchmark.

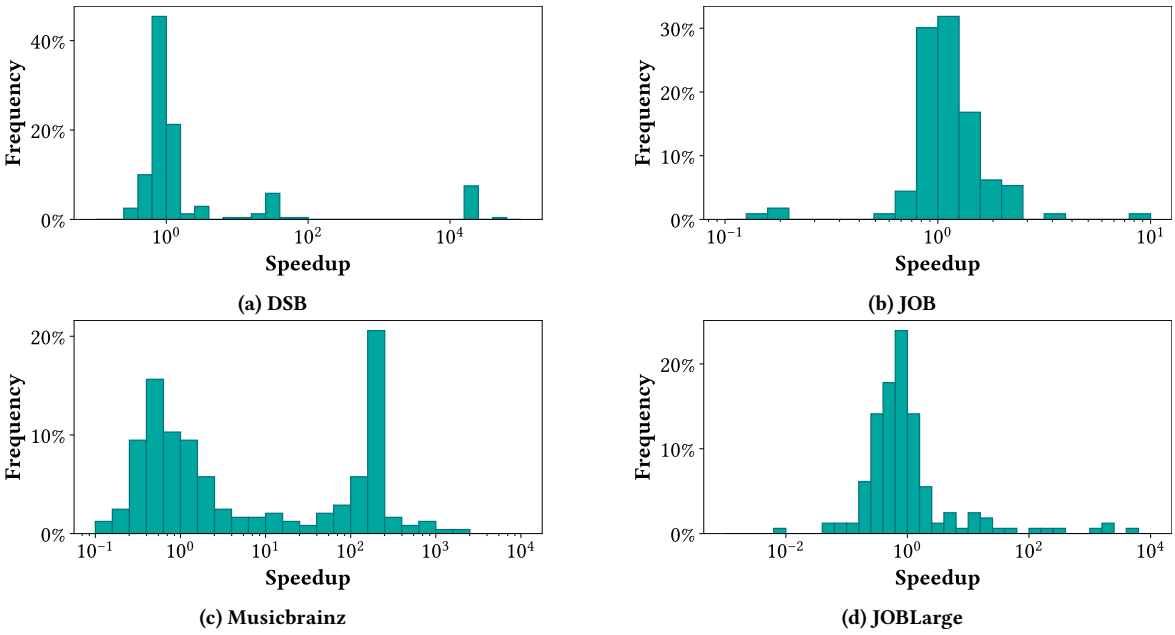
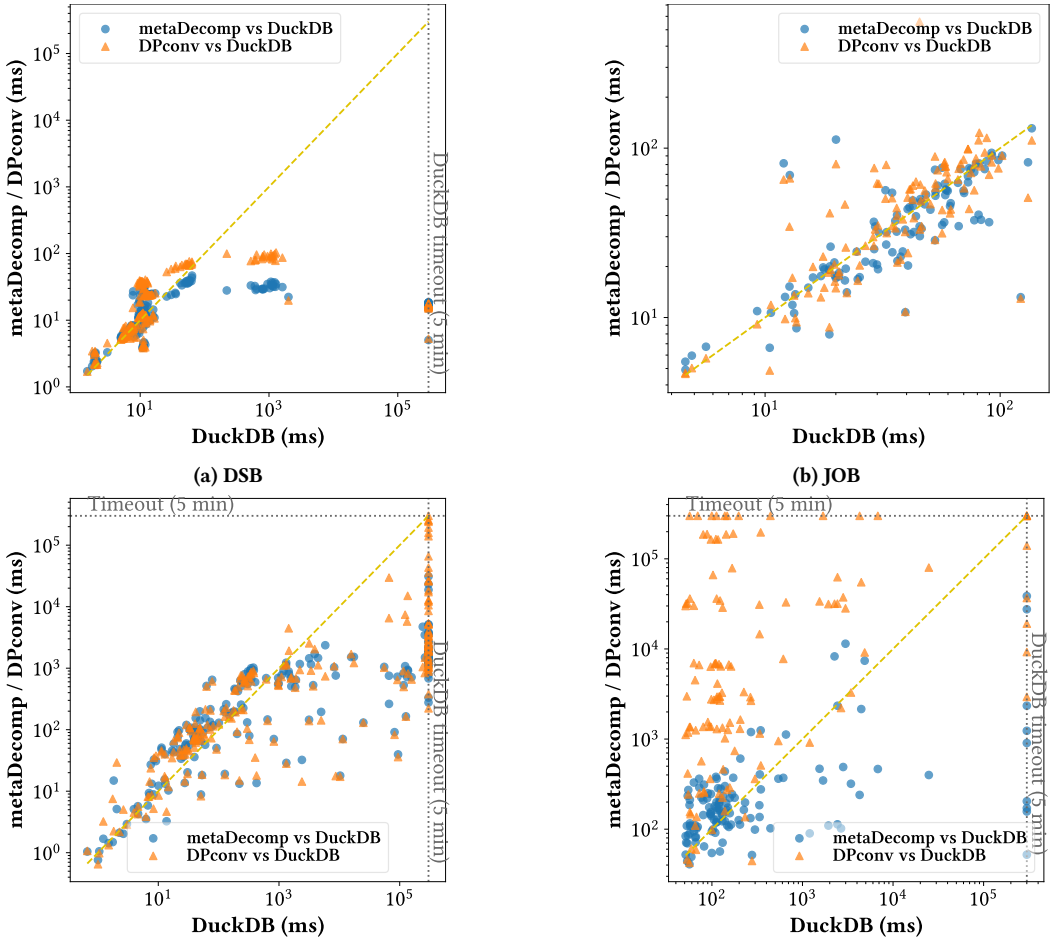


Figure 18: Histograms of the speedups of overall query evaluation time of metaDecomp over DuckDB on each benchmark

E.5 Total Query Evaluation Time Using metaDecomp/DPconv versus Using DuckDB on Each Benchmark

Fig. 19 shows the detailed breakdown of overall query evaluation time using metaDecomp and using DPconv measured for queries in each of the four benchmarks used in the paper. With the exception of JOBLarge, all approaches have similar performance for queries that takes DuckDB less than 100 ms. For those taking DuckDB more than 10 ms, both DPconv and metaDecomp start to show significant speedups. The relative speedups between DPconv and metaDecomp are shown by Section 6.2.3 and Fig. 17, where metaDecomp outperforms DPconv starting from approximately 1 second. On JOBLarge, we can already see DPconv being significantly slower than both DuckDB and metaDecomp for queries that takes DuckDB and metaDecomp less than 100 ms, mostly due to the sheer number of relations being joined, regardless of the required execution time.



(a) DSB (b) JOB (c) Musicbrainz (d) JOBLarge

Figure 19: Scatter plot of overall query evaluation time of using metaDecomp/DPconv versus using DuckDB.